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# The *UBE3A*-ATS antisense oligonucleotide rugonersen in children with Angelman syndrome: a phase 1 trial

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**Jörg F. Hipp[1]✉, Carlos A. Bacino[2], Lynne M. Bird[3], Ina Bruenig-Traebert[1], Daniel Chan[4], Marie Claire de Wit[5], Paulo Fontoura[6], Gregory Hooper[7], Ravi Jagasia[1], Michelle L. Krishnan[1], Lorraine Murtagh[1], Alessandro Noci[8], Ana Roche Martínez[9], Dietmar Schwab[10], Mercedes Serrano[11], Mark D. Shen[12], Julian Tillmann[1], Jorrit Tjeertes[8], Brenda Vincenzi[8], Elizabeth Berry-Kravis[13,24], Azad Bonni[1,24], the Rugonersen Study Group* & all TANGELO investigators***

Angelman syndrome (AS) is a severe genetic neurodevelopmental disorder with no disease-modifying treatments. AS is caused by deletion or mutation of the neuronally imprinted gene encoding the ubiquitin-protein ligase E3A (UBE3A). Rugonersen (RO7248824) is an antisense oligonucleotide that reinstates *UBE3A* by derepressing the silenced paternal allele. TANGELO was a phase 1, multicenter, open-label, multiple-ascending-dose trial with a long-term extension to investigate the safety and tolerability (primary) and pharmacokinetics (secondary) of rugonersen in children aged 1–12 years with AS ($n = 61$, F/M: 28/33). Key exploratory endpoints assessing changes following rugonersen treatment were electroencephalogram $\delta$-power (2–4 Hz) and domains of the Bayley Scales of Infant and Toddler Development—Third Edition and Vineland Adaptive Behavior Scales—Third Edition. The primary endpoint was met; rugonersen had an acceptable safety and tolerability profile. Analysis of exploratory endpoints showed that rugonersen led to a dose-dependent partial normalization of the AS-associated electroencephalogram abnormality and revealed signals of clinical improvement in core AS symptom domains beyond expectation from natural history data. The results of the primary study objective support continued development of rugonersen for AS. ClinicalTrials.gov registration: NCT04428281.

Angelman syndrome (AS) is a rare genetic neurodevelopmental disorder occurring in approximately 1 in 20,000 births[1–3] and with no available disease-modifying treatments. Clinical characteristics of AS include global developmental delay, intellectual disability, lack of speech, epilepsy, abnormal electroencephalogram (EEG), sleep difficulties, and characteristic dysmorphisms and behavioral features[4–9].

AS is caused by loss of function of the maternally inherited allele of the gene encoding ubiquitin-protein ligase E3A (UBE3A) located on chromosome 15q11.2–q13.1 (refs. 4,10–13). *UBE3A* is expressed biallelically in all cells, except neurons, where it is paternally imprinted (expressed only from the maternal allele), and has an important role in neuronal development and function[10,14,15]. Several genetic mechanisms cause AS including deletions of the maternal 15q11.2–q13.1 region that includes *UBE3A* (Del, ~70%), pathogenic variants of the maternal copy of *UBE3A* (Mut, ~15%), imprinting defects and paternal uniparental disomy of chromosome 15 (both ~5%).

A full list of affiliations appears at the end of the paper. ✉e-mail: joerg.hipp@roche.com

The imprinting of *UBE3A* in neurons is mediated by a paternal *cis*-acting long non-coding antisense transcript (*UBE3A*-ATS) that interferes with the transcription of *UBE3A* from the paternal allele[16–18]. Interrupting transcription of *UBE3A*-ATS, thereby allowing expression of *UBE3A* from the intact but silenced paternal allele, has been proposed as a possible strategy to treat AS[19,20]. This approach is supported by a substantial body of preclinical evidence in AS induced pluripotent stem cell (iPSC)-derived neurons and rodent models of AS, which includes pharmacological cleavage of *UBE3A*-ATS/*Ube3a*-ats (small molecules and antisense oligonucleotides)[19–23]. These studies and induction of UBE3A in AS mice using a Cre-dependent genetic approach[24] show that reinstatement of *Ube3a* is accompanied by partial normalization of EEG abnormalities; improvements of distinct phenotypes including open field, marble burying, nest building and forced swim test behaviors, motor performance on the rotarod and epilepsy; and enhanced hippocampal plasticity in slice preparations. In mice, the degree of improvement is largest with the youngest age of treatment onset, although hippocampal plasticity still improves with treatment beginning in adulthood. In sum, there is strong preclinical support for translating *UBE3A*-ATS-targeting therapies into the clinic.

Rugonersen (RO7248824) is a highly potent *UBE3A*-ATS-targeting ASO with locked nucleic acid backbone modification that was discovered in a screen using neurons derived from patients with AS[25]. Rugonersen has shown long-lasting pharmacodynamic effects with reduction of *Ube3a*-ats levels together with increased *Ube3a* mRNA and UBE3A protein when administered intrathecally in nonhuman primates (NHPs)[25]. Preclinical toxicologic studies showed an acceptable safety profile that supported clinical studies.

Here we report on the first-in-human study of rugonersen to assess safety, tolerability and pharmacokinetics (PK) in children 1–12 years of age with AS of Del and Mut genotypes (TANGELO; NCT04428281). The study has several noteworthy design elements. First, the age range reflects a carefully considered risk–benefit tradeoff anticipating higher benefits of treatment when started at younger ages. Second, the study implemented a within-participant dose escalation to explore various dose levels within the scope of a phase 1 multiple ascending dose (MAD) phase. Following the MAD phase, a long-term extension (LTE) phase gathered further safety and tolerability data. Furthermore, a 'bridging dose' was implemented aimed at reducing gaps between drug administration for participants enrolled in the MAD phase before implementation of the LTE phase.

To investigate potential effects of rugonersen on brain function, EEG was performed as an exploratory assessment at several timepoints. Natural history (NH) studies have shown that AS is characterized by EEG abnormalities that are particularly pronounced in the lower frequency range ($\delta$-frequency range, defined here as 2–4 Hz)[8,9]. EEG $\delta$-power has been systematically studied as a quantitative biomarker of AS pathophysiology. In AS, EEG $\delta$-power is strongly increased compared with age-matched typically developing individuals[26,27], present for both Del and Mut AS, suggesting that it relates to neuronal UBE3A loss[27], and has been shown to correlate cross-sectionally and longitudinally with symptom severity[28,29]. EEG $\delta$-power is also elevated in rodent models of AS[26] and was partially normalized in response to an ASO targeting *Ube3a*-ats (ref. 23). In sum, EEG $\delta$-power provides a candidate response biomarker to reflect changes in underlying abnormal brain function as expression of *UBE3A* is modified.

To assess the exploratory efficacy of rugonersen on core symptoms of AS, including cognition, receptive and expressive communication, and fine and gross motor function, we used three complementary instruments: the Bayley Scales of Infant Development—Third Edition (BSID)[30], an individually administered performance outcome assessment; the Vineland Adaptive Behavior Scales—Third Edition (VABS)[31], a semi-structured interview with the caregiver; and the Symptoms of Angelman Syndrome—Clinician Global Impression (SAS-CGI)[32], a clinician-rated assessment of severity (SAS-CGI-Severity) and symptom change (SAS-CGI-Change).

For the analysis of EEG $\delta$-power, BSID and VABS, we leveraged NH data (AS Natural History Study NCT00296764 (refs. 27,33); new AS-NHS, NCT04507997; FREESIAS[34]) to model and account for expected changes due to brain maturation and development over the duration of the study. Furthermore, we estimated minimal clinically important group differences and the cross-sectional relationship between EEG $\delta$-power and clinical scores from the NH data to provide context for the observed changes in clinical scales and EEG $\delta$-power.

Here we show that rugonersen had an acceptable safety and tolerability profile, led to a dose-dependent partial normalization of the characteristic EEG abnormality and showed signals of consistent clinical improvement deviating from the NH trajectory of the syndrome.

## Results

### Study design and participants

The phase I study, TANGELO (NCT04428281), is a multicenter, open-label, adaptive, MAD, intra-patient dose escalation study with an LTE and optional open-label extension (OOE), with the primary objective of investigating the safety and tolerability of rugonersen in patients with AS aged 1–12 years. Between August 2020 and March 2022, 50 participants with Del or Mut AS were enrolled at 12 sites in the United States, Italy, Spain and the Netherlands, into 8 MAD cohorts (Fig. 1, Table 1 and Methods; first and last patient enrolled: 17 August 2020 and 23 March 2022; 69 screened across all study parts). All participants received two or three intrathecally (IT) administered doses of rugonersen, ranging from 6 mg to 180 mg, within the first 2 months. The dosing period was followed by a posttreatment period that included two in-clinic visits on days 100 and 224. Doses were escalated within participants and across cohorts, with the older A cohorts (5–12 years) preceding the younger B cohorts (1–4 years).

Following the MAD, the LTE part was initiated upon supportive data coming from the 39-week chronic toxicity study in NHPs (some early cohorts received one 'bridging dose' while waiting for the initiation of the LTE). A total of 49 participants transitioned into the LTE, which started in May 2022 and ended in September 2024 (first and last patient enrolled: 21 April 2022 and 7 November 2022). Participants from MAD cohorts A1, A2, B1 and B2 were treated with a dose regimen of 120 mg every 16 weeks (Q16w-120mg, EA2 and EB2). Participants from MAD cohorts A3, A4, B3 and B4 were treated with a dose regimen of 180 mg every 24 weeks (Q24w-180mg, EA3 and EB3). Furthermore, 11 new participants were recruited to receive a dose regimen of 60 mg every 16 weeks (Q16w-60mg; EA1 and EB1; Table 1). Clinic visits with comprehensive assessments were performed the day before the IT injections. Following the LTE, the OOE with a reduced schedule of assessments was initiated and is still ongoing at the time of writing of this paper.

### Safety and tolerability

At the cutoff date of 23 September 2024 (end of the LTE), a total of 450 IT administrations of rugonersen have been performed in 61 participants, with some participants in the study for 4 years. A summary of the adverse events is presented in Table 2. Rugonersen was safely administered up to a dose of 180 mg, and no participants were withdrawn from the study owing to adverse events (AEs). Serious adverse events (SAEs) were experienced by 34% of participants ($n = 21$). SAEs were classified as drug related by the investigators for 13% of participants ($n = 8$). The drug-related SAEs included six grade 3 (severe) events classified as seizure ($n = 2$), epilepsy ($n = 1$), vomiting ($n = 1$), pyrexia ($n = 1$) and post-lumbar puncture syndrome ($n = 1$). The remaining three drug-related SAEs were grade 2 (moderate): procedural headache, irritability and cerebrospinal fluid (CSF) pressure increase (all $n = 1$). The treatment-related SAE of pyrexia was considered as a dose-limiting toxicity, which occurred after a cohort A1 participant received the 180 mg

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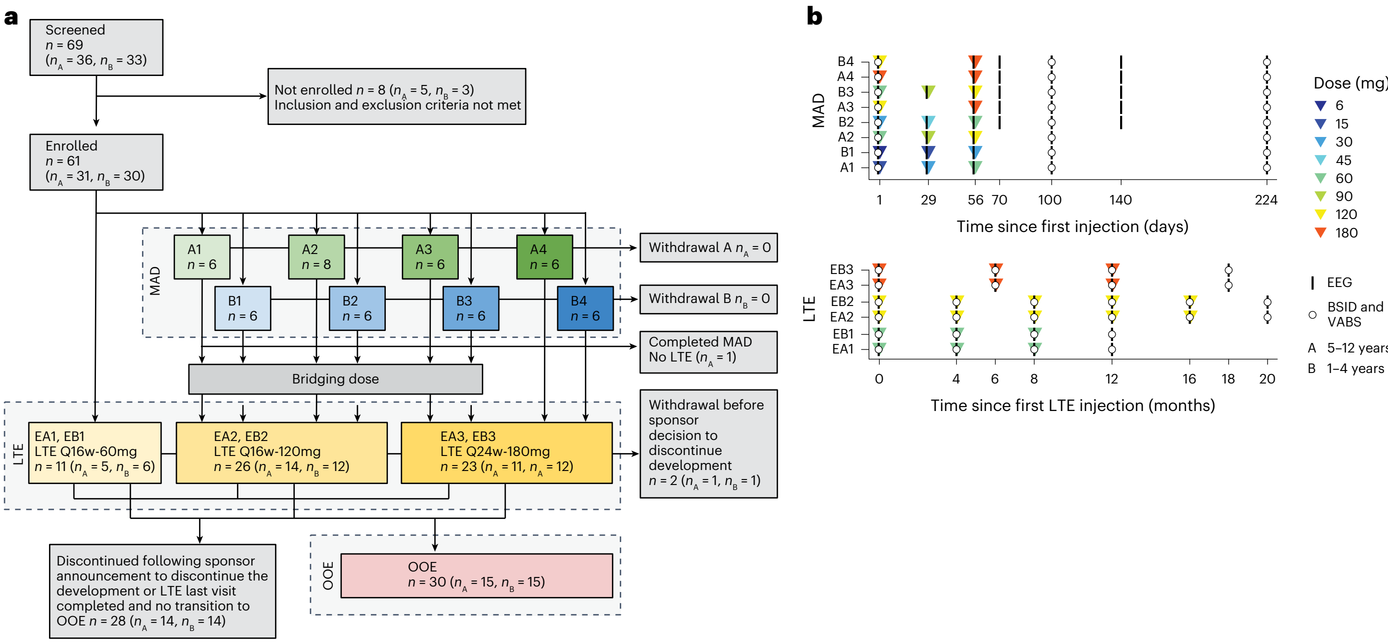


**Fig. 1 | Trial design and patient flow diagram. a**, Patient disposition through the MAD, LTE and OOE parts of TANGELO. **b**, Dosing schedules for MAD and LTE, respectively, and concurrent EEG and clinical assessments (BSID, VABS).

bridging dose 292 days after receiving 60 mg as the last MAD dose. The onset of pyrexia was 1 day after dosing, with a temperature exceeding 40 °C, which led to hospitalization 3 days after dosing. Blood cultures, urinalysis and viral PCR were negative, and magnetic resonance imaging (MRI) of the brain and spinal cord was normal. Pyrexia was treated with acetaminophen/paracetamol, ceftriaxone, acyclovir and ibuprofen and resolved with hospital discharge 13 days after the dose. The total number of AEs reported was 1,225, of which 503 were considered related to rugonersen. The most common AE was pyrexia, which was observed a few days after dosing and resolved without sequelae, within 2–5 days. Medication, for example, acetaminophen/paracetamol, was used for treatment, where indicated. The incidence and duration of pyrexia events showed a dose dependency with both frequency and duration increasing with increased dose. Other noteworthy AEs included vomiting, ataxia (most often reported as increased ataxia) and seizures. There were no events above grade 3. Overall, AE frequency was similar across different doses with numerically more and longer pyrexia events for 180 mg versus 120 mg in A cohorts (Supplementary Table 3). Across all cohorts for laboratory parameters, ECG and vital sign measurements, there were no clinically significant changes or trends identified (Supplementary Section 4).

Of 13 participants entering the TANGELO trial without history of epilepsy (10 Del, 3 Mut; mean age (s.d.): 4.3 (3.4) years; range: 1.8–12.8 years), 9 had onset of epilepsy during the study. The onset of epilepsy in these participants comported with expectation from NH data[35].

### Pharmacokinetics

Rugonersen plasma concentration reached the highest levels between 3 h and 6 h post-dose and was approximately dose proportional. Rugonersen concentrations in CSF samples, which were taken 1 month or later following IT administration, poorly reflected the dose levels. Both plasma and CSF rugonersen concentrations were of limited utility to infer brain concentration. See Supplementary Section 5 for more information on pharmacokinetics.

### Pharmacodynamics—EEG $\delta$-power

Awake in-clinic EEG assessments of 20 min duration were performed in the MAD and LTE phases to (1) determine whether rugonersen led to changes in pathological brain activity (EEG $\delta$-power) and (2) to gain insights into the dynamics and dose dependence of such a putative response to inform design of subsequent studies.

Baseline EEG $\delta$-power values of the study participants were within the expected range known from AS NH data (Supplementary Fig. 2b). EEG $\delta$-power decreases with chronological age in AS[27]. Thus, changes are expected over the study duration, independent of any potential treatment effects, simply due to maturation. To account for this age-related effect, we developed a statistical model that predicted age- and genotype-matched expected changes from NH data (Methods).

Using linear models, we investigated the change in EEG $\delta$-power relative to baseline (Fig. 2a and Supplementary Table 10). First, we investigated the average response across all age and dose groups. On day 55, following one or two doses of rugonersen, EEG $\delta$-power significantly decreased below the AS NH reference (−1.42 dB, corresponding to −27.9% change in signal power, $P = 0.004$, one-tailed $t$-test for decrease). The effect of rugonersen on the EEG $\delta$-power was more pronounced on day 100 (that is, approximately 6 weeks after the last MAD dose; −2.3 dB, corresponding to −41.3% change in signal power, $P = 1.64 \times 10^{-6}$, one-tailed $t$-test for decrease) and returned toward the NH reference on day 224 (that is, approximately 24 weeks after the last MAD dose; −1.03 dB, corresponding to −21.1% in signal power, $P = 0.014$, one-tailed $t$-test for decrease). These results show a dynamic change in brain electrical activity in response to rugonersen.

For day 100, the timepoint with the strongest EEG $\delta$-power effect, we investigated the EEG response to rugonersen in more detail (Supplementary Table 11). EEG $\delta$-power decreased in the direction of normalization (Fig. 2b) and was dose dependent (Fig. 2c; Pearson correlation coefficient between dose and EEG: $r = -0.59$, $P = 1.2 \times 10^{-5}$, two-tailed test). The EEG $\delta$-power decrease was found for all electrodes with an anterior–posterior gradient (Fig. 2d). The EEG spectral change was most pronounced in the $\delta$-frequency range and extended into neighboring frequencies (Fig. 2e). Furthermore, the EEG $\delta$-power decrease was more pronounced for older than for younger participants (Fig. 2f; Pearson correlation coefficient between age and EEG, $r = -0.39$, $P = 0.0047$, two-tailed test).

The data showed an EEG $\delta$-power decrease, exceeding the model-predicted NH reference, ranging from −1.61 dB for the younger

**Table 1 | Participant demographics and baseline characteristics**

| Characteristics | A cohorts (ages 5–12) | | | | | B cohorts (ages 1–4) | | | | | All |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | A1 | A2 | A3 | A4 | EA1 | B1 | B2 | B3 | B4 | EB1 | |
| | (*n*=6) | (*n*=8) | (*n*=7) | (*n*=5) | (*n*=5) | (*n*=6) | (*n*=6) | (*n*=6) | (*n*=6) | (*n*=6) | (*n*=61) |
| Mean age (s.d.) (years) | 9.8 (2.5) | 8.3 (2.6) | 9.2 (2.6) | 7.2 (2.8) | 9.2 (3.0) | 2.6 (0.9) | 3.6 (1.0) | 3.3 (1.0) | 3.1 (0.8) | 2.0 (1.3) | 5.4 (3.5) |
| Female (*n* (%)) | 3 (50.0%) | 3 (37.5%) | 3 (42.9%) | 3 (60.0%) | 3 (60%) | 2 (33.3%) | 3 (50.0%) | 5 (83.3%) | 2 (33.3%) | 4 (66.7%) | 28 (46%) |
| Race (white; *n* (%)) | 6 (100%) | 6 (75.0%) | 7 (100%) | 4 (80.0%) | 5 (100%) | 5 (83.3%) | 6 (100%) | 6 (100%) | 5 (83.3%) | 6 (100%) | 56 (92%) |
| Mean height (s.d.) (cm) | 132.8 (11.9) | 122.7 (11.8) | 131.6 (17.3) | 115.0 (14.7) | 92.5 (9.3) | 97.8 (9.8) | 93.5 (6.4) | 92.0 (5.6) | 133.8 (14.3) | 93.6 (11.9) | 110.8 (20.7) |
| Mean weight (s.d.) (kg) | 35.4 (9.4) | 26.1 (9.7) | 34.8 (14.2) | 22.6 (6.3) | 13.1 (2.5) | 16.3 (6.6) | 13.6 (3.2) | 13.0 (1.4) | 31.5 (7.1) | 13.8 (3.6) | 22.2 (11.4) |
| Mean head circumference (s.d.) (cm) | 51.6 (2.0) | 50.4 (1.9) | 51.6 (1.4) | 49.5 (1.4) | 46.5 (1.8) | 47.5 (1.6) | 46.3 (1.4) | 45.8 (2.5) | 50.1 (1.4) | 46.9 (1.7) | 48.7 (2.7) |
| **Genotype (*n* (%))** | | | | | | | | | | | |
| Class 1 deletion | 0 (0.0%) | 4 (50.0%) | 2 (28.6%) | 2 (40.0%) | 2 (40%) | 3 (50.0%) | 2 (33.3%) | 2 (33.3%) | 3 (50.0%) | 4 (66.7%) | 22 (36%) |
| Class 2 deletion | 4 (66.7%) | 2 (25.0%) | 3 (42.9%) | 3 (60.0%) | 3 (60%) | 3 (50.0%) | 3 (50.0%) | 3 (50.0%) | 2 (33.3%) | 2 (33.3%) | 28 (46%) |
| Atypical deletion | 1 (16.7%) | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0 | 1 (2%) |
| *UBE3A* mutation | 1 (16.7%) | 2 (25.0%) | 2 (28.6%) | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 1 (16.7%) | 1 (16.7%) | 1 (16.7%) | 0 | 8 (13%) |
| **BSID (GSV) (s.d.)** | | | | | | | | | | | |
| Cognitive | 550.0 (11.4) | 531.4 (29.9) | 523.9 (52.8) | 482.0 (22.6) | 487.8 (43.9) | 502.8 (30.8) | 491.8 (26.2) | 512.6 (28.1) | 530.0 (16.6) | 479.2 (22.0) | 508.6 (36.9) |
| Receptive communication | 516.5 (28.9) | 494.2 (44.1) | 513.3 (46.8) | 452.8 (34.7) | 477.3 (41.8) | 487.8 (24.5) | 470.2 (23.3) | 499.0 (21.0) | 514.5 (36.5) | 480.0 (30.1) | 490.2 (37.5) |
| Expressive communication | 473.2 (38.0) | 451.2 (36.0) | 462.7 (31.3) | 419.4 (39.2) | 420.3 (21.2) | 438.3 (13.4) | 432.8 (40.7) | 435.2 (22.2) | 441.8 (8.6) | 430.6 (16.1) | 440.7 (31.1) |
| Fine motor | 544.5 (10.8) | 513.6 (31.7) | 529.4 (54.5) | 469.2 (20.7) | 478.5 (43.5) | 478.7 (15.6) | 476.5 (24.5) | 482.8 (52.0) | 514.0 (37.3) | 479.0 (33.1) | 496.6 (41.3) |
| Gross motor | 568.0 (17.4) | 549.0 (26.1) | 545.1 (45.6) | 505.0 (33.4) | 482.7 (47.7) | 505.0 (40.5) | 490.5 (45.6) | 508.0 (26.0) | 570.2 (11.2) | 473.0 (27.8) | 519.6 (46.0) |
| **VABS (GSV) (s.d.)** | | | | | | | | | | | |
| DLS—personal | 90.3 (8.6) | 85.1 (6.8) | 91.3 (15.0) | 72.6 (5.5) | 66.0 (19.0) | 63.7 (11.0) | 63.2 (25.4) | 76.3 (13.8) | 91.0 (11.8) | 68.5 (5.0) | 77.2 (17.0) |
| Receptive communication | 92.5 (16.9) | 81.2 (11.3) | 84.1 (14.7) | 55.6 (18.5) | 71.3 (14.0) | 65.7 (12.7) | 74.2 (6.0) | 75.2 (4.9) | 90.2 (7.2) | 71.8 (7.6) | 76.5 (15.4) |
| Expressive communication | 88.7 (12.2) | 87.9 (13.1) | 82.1 (11.1) | 71.2 (7.0) | 69.7 (20.7) | 76.0 (10.4) | 69.7 (16.7) | 75.8 (22.3) | 97.2 (11.4) | 64.8 (6.1) | 78.6 (16.1) |
| Fine motor | 108.5 (10.3) | 96.2 (5.0) | 103.9 (20.9) | 75.4 (16.2) | 77.0 (15.0) | 84.8 (10.9) | 80.8 (12.8) | 84.5 (15.4) | 105.2 (7.5) | 73.5 (1.7) | 89.7 (17.2) |
| Gross motor | 127.5 (21.5) | 117.9 (12.5) | 120.9 (26.9) | 100.4 (18.6) | 86.7 (28.0) | 92.7 (15.9) | 85.8 (33.4) | 99.0 (14.5) | 130.8 (9.4) | 86.8 (3.0) | 105.4 (25.4) |
| **SAS-CGI-Severity (s.d.)** | | | | | | | | | | | |
| Seizures | 1.3 (1.2) | 1.4 (1.1) | 0.4 (0.8) | 1.4 (0.5) | 1.2 (1.5) | 1.2 (1.0) | 0.8 (1.0) | 1.0 (1.3) | 1.2 (0.8) | 1.4 (1.3) | 1.1 (1.0) |
| Sleep | 2.0 (1.1) | 1.6 (1.6) | 2.4 (1.5) | 1.4 (1.1) | 2.5 (0.8) | 1.5 (1.6) | 1.2 (1.0) | 1.5 (1.0) | 2.2 (1.3) | 3.0 (0.7) | 1.9 (1.3) |
| Maladaptive behavior | 2.5 (0.5) | 2.5 (1.2) | 3.0 (1.4) | 2.0 (1.2) | 1.8 (0.8) | 1.7 (0.8) | 2.2 (0.8) | 1.5 (1.2) | 2.6 (1.8) | 0.8 (0.8) | 2.1 (1.2) |
| Expressive communication | 3.3 (0.5) | 2.8 (1.3) | 3.0 (1.0) | 4.0 (0.7) | 3.2 (1.0) | 2.8 (1.0) | 2.8 (0.4) | 2.8 (0.8) | 3.4 (0.5) | 3.4 (1.3) | 3.1 (0.9) |
| Fine motor | 2.8 (0.4) | 2.6 (0.7) | 2.6 (0.5) | 3.2 (0.8) | 2.7 (0.8) | 2.7 (0.5) | 2.3 (0.8) | 2.2 (0.4) | 2.6 (1.1) | 3.6 (1.1) | 2.7 (0.8) |
| Gross motor | 2.3 (0.8) | 2.2 (0.9) | 2.4 (1.3) | 3.4 (0.5) | 2.8 (0.8) | 2.5 (0.5) | 2.7 (0.5) | 2.8 (1.0) | 2.2 (0.8) | 3.4 (0.5) | 2.6 (0.9) |
| Cognitive | 3.0 (0.9) | 2.9 (0.8) | 3.1 (1.1) | 3.6 (0.5) | 2.8 (0.8) | 3.0 (0.6) | 3.0 (0.6) | 2.7 (0.8) | 2.6 (0.5) | 3.4 (1.1) | 3.0 (0.8) |
| DLS | 3.2 (1.0) | 3.5 (0.5) | 3.1 (1.1) | 3.8 (0.4) | 2.7 (1.6) | 3.2 (0.8) | 3.3 (0.8) | 3.2 (0.8) | 3.0 (1.0) | 3.0 (1.9) | 3.2 (1.0) |
| Overall | 2.8 (0.8) | 2.9 (0.6) | 2.9 (0.7) | 3.6 (0.5) | 3.0 (0.6) | 2.8 (0.4) | 2.8 (0.4) | 2.3 (0.5) | 2.8 (0.8) | 3.0 (0.0) | 2.9 (0.6) |

Demographics and baseline characteristics (VABS, BSID and SAS-CGI-S) for MAD cohorts B1–B4 and A1–A4, LTE cohorts EA1 and EB1, and the summary across all cohorts (see Supplementary Table 2 for baseline values for additional exploratory clinical scales). For the clinical scales, numbers were in part lower than the cohort numbers reported in line 2 owing to few assessments that were not performed or were invalid. DLS, daily living skills.

age group and a cumulative MAD dose of 240 mg to −6.03 dB for the older age group and a cumulative MAD dose of 320 mg (Supplementary Table 12; corresponding to −31.0% and −75.1% change in signal power, respectively). Using the cross-sectional relationship between EEG $\delta$-power and clinical scales from NH data (Supplementary Section 10) to translate changes in EEG $\delta$-power to changes in clinical scores (BSID), we found that the changes on the order of −4 dB correspond to changes in BSID that we estimated to be minimal clinically important group differences, while −6 dB surpassed the estimate (Supplementary Section 8).

Throughout the LTE, the group on the Q16w-120mg dose regimen showed a persistent decrease of EEG $\delta$-power below the NH reference, even though EEGs were performed at times when rugonersen concentration was at its trough (that is, the day before next dosing; average across LTE visits 2–6: −2.31 dB, corresponding to −41% decrease in signal power; Fig. 2g, left, and Supplementary Table 15). Spatial and spectral distribution were largely in line with the MAD results, with the strongest effects in the $\delta$-frequency range and an anterior–posterior gradient (Supplementary Fig. 6a–c). We found a clear dose-regimen dependence, with Q16w-120mg showing a persistent separation from the NH reference, and no clear separation for the Q24w-180mg and Q16w-60mg dose (Fig. 2g, center and right).

For additional MAD and LTE EEG analyses, including analyses of alternative EEG endpoints, sex and genotype, see Supplementary Sections 12 and 13.

**Table 2 | Adverse event summary table**

| | All participants | Grade 1 (mild) | Grade 2 (moderate) | Grade 3 (severe) |
|---|---|---|---|---|
| Number of participants with at least one AE | 59 (96.7%) | 5 (8.2%) | 22 (36.1%) | 32 (52.5%) |
| Number of participants with at least one SAE | 21 (34.4%) | 1 (1.6%) | 3 (4.9%) | 17 (27.9%) |
| Number of participants with at least one drug-related SAE | 8 (13.1%) | 0 (0.0%) | 2 (3.3%) | 6 (9.8%) |
| **Most common AEs (over 20%)** | | | | |
| Pyrexia | 52 (85.2%) | 19 (31.1%) | 21 (34.4%) | 12 (19.7%) |
| Vomiting | 41 (67.2%) | 23 (37.7%) | 14 (23.0%) | 4 (6.6%) |
| Ataxia | 25 (41.0%) | 11 (18.0%) | 13 (21.3%) | 1 (1.6%) |
| Covid-19 | 20 (32.8%) | 16 26.2%) | 3 (4.9%) | 1 (1.6%) |
| Seizure | 19 (31.1%) | 5 (8.2%) | 9 (14.8%) | 5 (8.2%) |
| CSF pressure increased | 16 (26.2%) | 13 21.3%) | 1 (1.6%) | 2 (3.3%) |
| Headache | 16 (26.2%) | 10 (16.4%) | 6 (9.8%) | 0 (0.0%) |
| Upper respiratory tract infection | 16 (26.2%) | 10 (16.4%) | 6 (9.8%) | 0 (0.0%) |
| Decreased appetite | 15 (24.6%) | 14 (23.0%) | 1 (1.6%) | 0 (0.0%) |
| Fatigue | 15 (24.6%) | 11 (18.0%) | 4 (6.6%) | 0 (0.0%) |
| Nausea | 14 (23.0%) | 8 (13.1%) | 6 (9.8%) | 0 (0.0%) |

Numbers and percentages of all participants. Grading is included for the highest National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) (v5.0) grade experienced by the participant for each group. One participant's grading is based on the initial grade as no most-extreme grade was reported.

### Exploratory clinical efficacy analyses

BSID and VABS baseline values fell within the range expected from NH data (Table 1 and Supplementary Fig. 3). As with the EEG analysis, we used AS NH data-derived models to account for developmental changes over the course of the study.

In the MAD, increases from baseline following dosing of rugonersen assessed at days 100 and 224 exceeded the NH reference for all five VABS subdomains and four of five BSID domains (for most comparisons, $P < 0.05$, one-tailed $t$-test for increases; Fig. 3a,b and Supplementary Table 17). Changes observed were comparable in magnitude of the derived estimate of minimal clinically important group differences for the BSID scales (Supplementary Section 8). The general improvement above NH was consistent with clinician ratings of change, with ≥74% having an improvement on the SAS-CGI-Change (Fig. 3c).

For day 224, the timepoint with the stronger clinical responses in the MAD, we performed additional analyses to test for dependence on age, genotype, sex, dose and the magnitude of the EEG response on day 100 (Supplementary Fig. 7 and Table 18). There were no associations with dose, genotype or age. There was a weak trend for individuals with a stronger EEG $\delta$-power response at day 100 to show larger clinical improvements (5 of 20 tests, $P < 0.1$, uncorrected; but 8 of 20 effects pointing into the unexpected direction, including one with $P < 0.1$, two-tailed $t$-test, uncorrected). Furthermore, there was a trend for females showing a larger response (three domains, $P < 0.1$, two-tailed $t$-test, uncorrected).

Consistent with the MAD results, the LTE showed increases above the NH reference in all VABS subdomains and 4 of 5 BSID domains (Fig. 4a,b and Supplementary Table 19), and improvements in severity relative to baseline as assessed by SAS-CGI-Severity (Supplementary Fig. 9). Changes observed in the LTE were generally of greater magnitude than the estimate of minimal clinically important group differences for the BSID (red dashed lines in Fig. 4a,c). All dose regimens (Q16w-120mg, Q24w-180mg, Q16w-60mg) showed similar magnitudes of separation from the NH reference on BSID and VABS domains (Fig. 4c,d and Supplementary Fig. 8c,f). Of note, a comparison between LTE cohorts is complicated by differences in assessment timepoints (4 months versus 6 months following the last dose) and in duration within the study at the start of LTE (EA1/EB1 new joiners for LTE, EA2/EB2 in the study for about one and a half years at the start of LTE).

For LTE visit 2, the latest LTE timepoint with data for most participants, we performed additional analyses to test for dependence on age, genotype, sex, dose and magnitude of the EEG effect at the same visit (Supplementary Fig. 8 and Table 20). There was no sign of dose, genotype or age dependence. There was a trend for individuals with a stronger EEG $\delta$-power response to show larger clinical improvement (4 of 20 tests, $P < 0.1$, two-sided $t$-test, uncorrected; 19 of 20 effects pointing into the expected direction). Furthermore, there was a trend for females showing a larger response (one domain, $P < 0.1$, two-sided $t$-test, uncorrected; same direction for all 10 scales).

## Discussion

Rugonersen had an acceptable safety and tolerability profile, led to a dose-dependent partial normalization of the characteristic EEG abnormality and was accompanied by signals of improvement in key clinical domains highly relevant to AS. The improvements exceeded the NH expected trajectory plus estimates of minimal clinically important group differences. These data provide encouraging clinical evidence for UBE3A-reinstating therapies in individuals with AS and represent a demonstration of altered brain activity in a neurodevelopmental disorder with a treatment targeting the genetic cause of the disease.

AEs and SAEs seen with rugonersen treatment generally comported with expectations for the drug class, route of administration (IT injection under sedation or anesthesia), patient population and study duration. Nausea and vomiting are common after general anesthesia in pediatric patients[36]. There is an expected event rate for post-lumbar puncture syndrome with IT administration, with headache occurring in up to one-third of participants[37–39]. Recognition of headaches can be challenging in nonverbal participants. Post-lumbar puncture syndrome was experienced by three participants (of which one event was considered related to rugonersen), and procedural headache was experienced by one participant. Although pyrexia is reported as a common adverse reaction to ASO molecules in more than 20% of patients[40], the incidence was high following rugonersen administration. The pyrexia was manageable and reversible through treatment with nonprescription drugs, for example, acetaminophen/paracetamol, as needed, and prophylaxis for pyrexia was implemented based on investigator preference and patient need. All events of pyrexia occurring post administration resolved. Epilepsy is very common in Del AS (close to 100% at age 5) and also frequent in Mut AS (~75% at age 10)[35]. In sum, rugonersen had an acceptable safety and tolerability profile.

EEG $\delta$-power is characteristically abnormal in AS and provides a macroscopic measure of pathological brain function caused by lack of UBE3A in neurons. Our results show that rugonersen partially normalized EEG $\delta$-power in a dose-dependent manner. EEG $\delta$-power was reduced by ~2.3 dB in the MAD on day 100 on average across all dose groups and in the LTE for the dose regimen Q16w-120mg at trough concentration. The magnitude of reduction was more substantial for higher doses around peak concentration assessed in the MAD (up to −6 dB), corresponding to a drop of ~75% in signal power. There was a trend for individuals with a stronger EEG response to show a larger clinical response, which was more pronounced in the LTE compared with the MAD. Furthermore, for the cross-sectional relationship between EEG $\delta$-power and clinical scales in NH data, we found that changes on the order of −4 dB correspond to changes in BSID that we estimated to be minimal clinically important group differences. In sum, these

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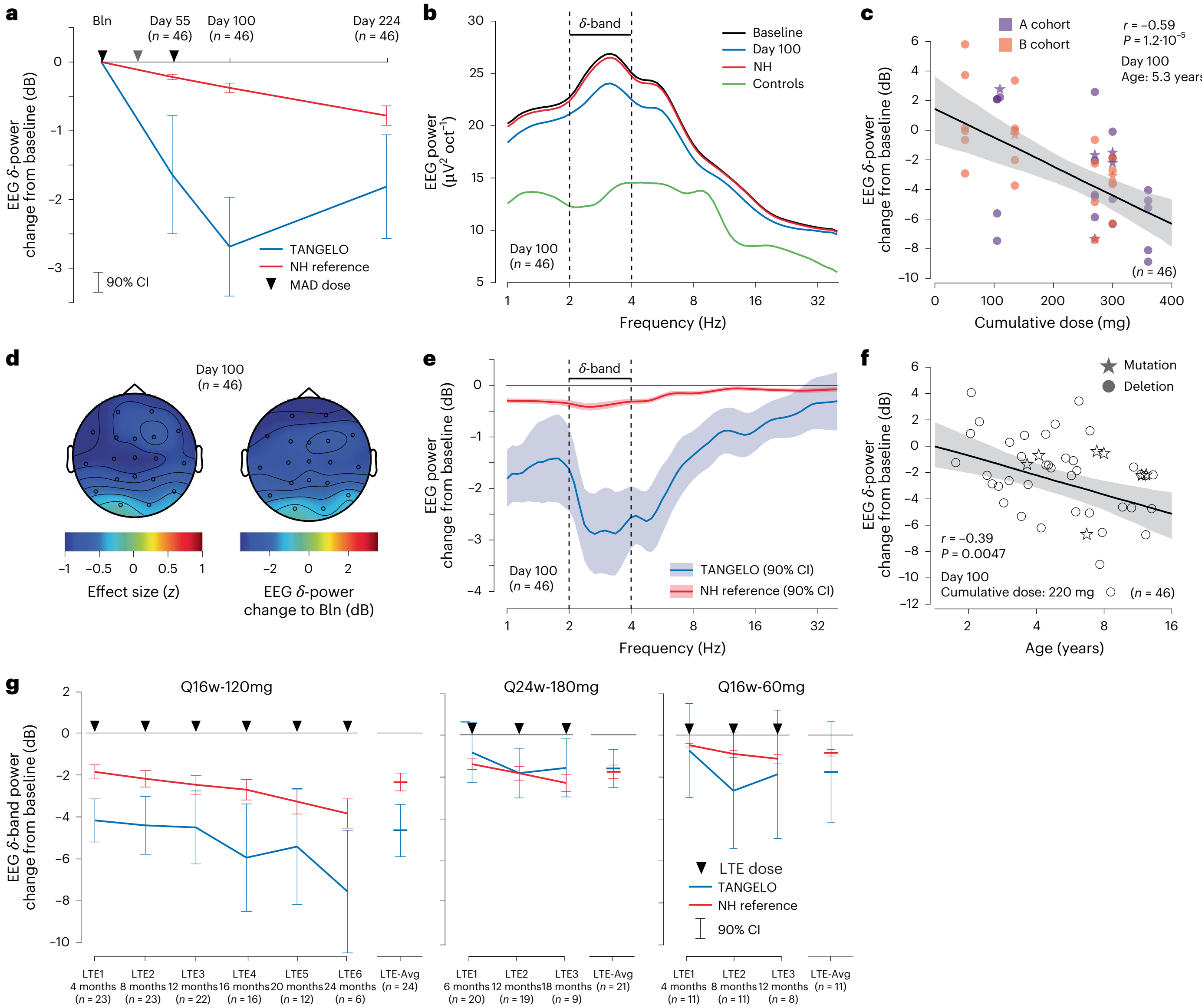


**Fig. 2 | EEG responses to rugonersen. a**, LS mean changes from baseline in EEG $\delta$-power (2–4 Hz) for MAD participants (blue) and age-matched NH predictions (red). Timepoints of rugonersen doses are shown by black triangles. **b**, LS mean power spectra. MAD participants at baseline (black) and day 100 (blue). Age- and genotype-matched AS NH predictions (red) and age-matched predictions for typically developing children (green). **c**, Correlation between cumulative dose (sum of all MAD doses for a participant) and change in EEG $\delta$-power on MAD day 100 from baseline predicted from the linear model (age set to mean log-age of 5.3 years). Age groups are color coded; AS genotypes are coded by shape (Del, circle; Mut, star). Line, linear fit; gray shade, 90% confidence interval (CI); Pearson correlation, two-sided test. **d**, Topography of LS mean EEG $\delta$-power on MAD day 100 change from baseline and accounting for predicted changes from NH. Left: effect size in $z$-scores; right: change in dB. **e**, LS mean power spectral change from baseline for MAD participants (blue) and age-matched NH predictions (red) on MAD day 100. **f**, Correlation between age at baseline and the individual change in EEG $\delta$-power on MAD day 100 relative to baseline predicted from the linear model (cumulative dose set to 220 mg). See **c** for more information. **g**, EEG responses in the LTE for all three LTE dose regimens. LS mean change from baseline in EEG $\delta$-power for participants (blue) and age-matched NH predictions (red). LTEi corresponds to the assessment following the $i$th LTE dose, right before the $(i+1)$th dose. LTE-Avg, average of available LTE timepoints > LTE1 for each participant. See Supplementary Figs. 5 and 6 for additional analyses.

data suggest that rugonersen, at tolerable doses, led to a substantial reduction of pathological EEG activity that may be related to clinically meaningful improvements.

The MAD design with two or three successive doses within 2 months, followed by a 6-month observation period, allowed the study of the dynamics of the EEG $\delta$-power response. The EEG $\delta$-power change reached a maximum around 2 months after the last MAD dose and strongly abated 6 months following the last MAD dose. Strikingly, this dynamic nature of the EEG response, both in terms of onset and fading of the effect, suggests that the changes in brain function closely mirrored the drug concentration. Alternatively, changes in EEG might have taken a much longer time to emerge or persist, once established. The finding of a dynamic EEG $\delta$-power response may have implications for the further development of rugonersen and, more generally, for UBE3A-reinstating therapies. It is plausible to assume that changes in brain function precede and underlie potential clinical improvements. Consequently, dose regimens that maximize the EEG $\delta$-power effect and avoid substantial reduction between doses are preferred. Support for this interpretation comes from reports from investigators in this study that improvements in such domains as attention, behavior and sleep were most pronounced around 2 months post-dose and had partly decayed by LTE visits 4–6 months after a dose. The sustained

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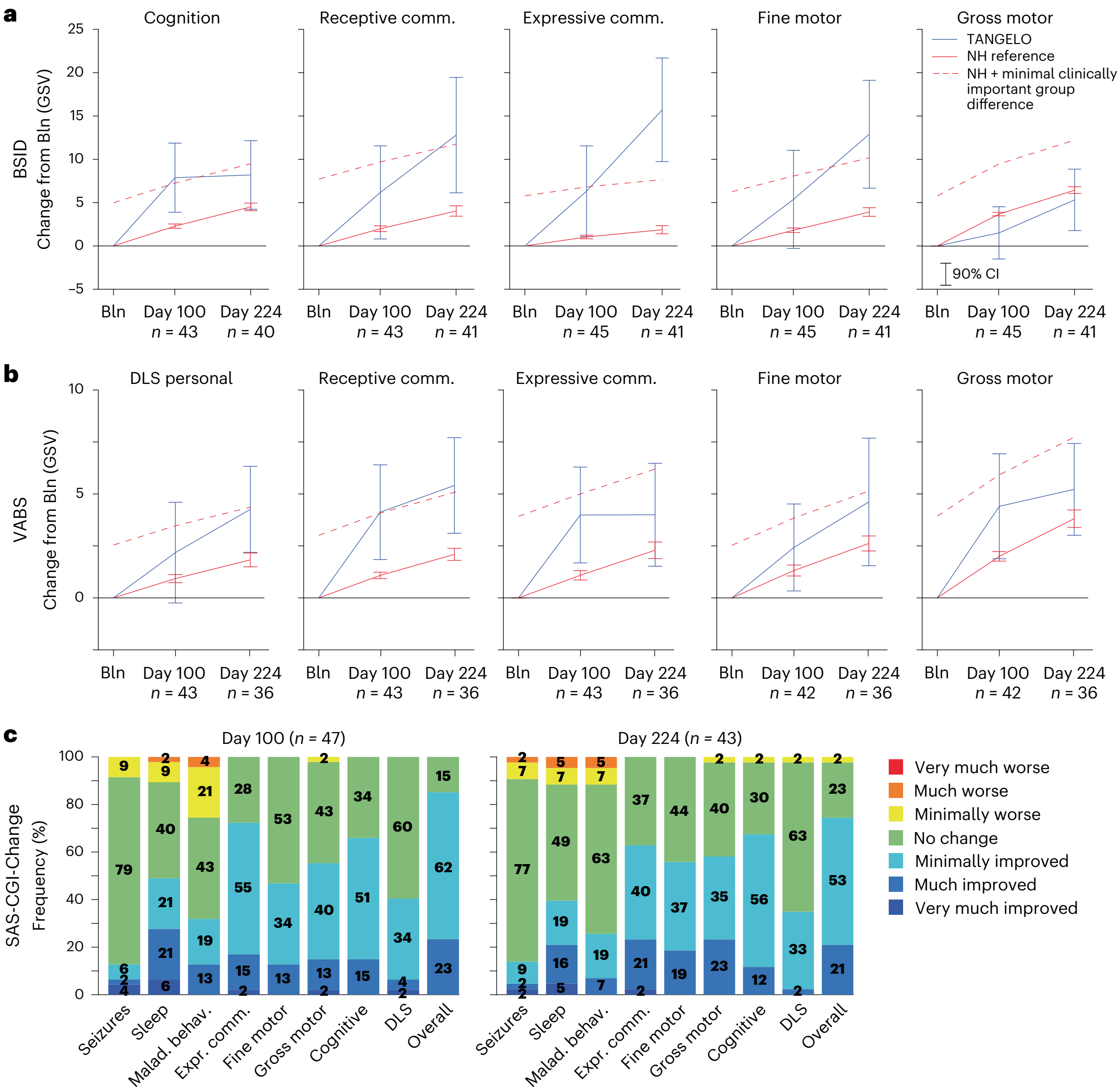


**Fig. 3 | Clinical response in the MAD. a**, BSID LS mean changes from baseline for MAD participants (blue) and the age-matched NH model prediction (red). The error bars show 90% CIs. The red dashed lines show an estimate of the minimal clinically important group difference on top of the NH model prediction (Supplementary Table 7). Positive changes reflect improvements. **b**, Same as **a** for VABS. **c**, SAS-CGI-Change for days 100 and 224. Malad. behav., maladaptive behavior; Receptive comm., receptive communication; Expr. comm., expressive communication.

EEG $\delta$-power effect observed for the Q16w-120mg dose regimen, but not the other dose regimens studied, provides strong support for the Q16w-120mg dose regimen for rugonersen. Furthermore, the totality of our data suggests that more frequent dosing of rugonersen (for example, Q12w), closer to the duration of the peak EEG response, is expected to produce a stronger sustained effect on EEG $\delta$-power and may hold the prospect of greater clinical improvement.

In the context of neurodevelopmental disorders, a disease-modifying treatment is expected to have the largest impact the earlier in life it is started[24]. Nevertheless, we found stronger EEG effects and numerically larger clinical improvements in the older (5–12 years) than in the younger (1–4 years) participants. There are several possible explanations for this observation. First, there could be age-related differences in drug delivery, CSF turnover and pharmacokinetics. Second, the need for UBE3A in neurons to support normal brain development and function may change with age. Furthermore, EEG $\delta$-power may reflect different neural processes at different ages, and a direct quantitative comparison between individuals of such different ages may not be valid. More work is needed to understand potential differences of the effect of UBE3A reinstatement in AS throughout development.

Our study provides new evidence for the utility of EEG $\delta$-power as a biomarker for the development of UBE3A-reinstating therapies in AS. EEG $\delta$-power can be used in early phase trials to show an impact on brain function and to support selection of dose amount and interval. Moreover, whereas the relationship between treatment-induced changes in EEG and clinical efficacy needs to be further investigated, EEG $\delta$-power has the potential as a surrogate endpoint biomarker for UBE3A-reinstating therapies in AS. Our findings may have implications for the use of quantitative EEG in other genetic disorders with characteristic EEG abnormalities, including Fragile X syndrome[41] (excess gamma-band power, a decrease and shift in alpha-band power), Dup15q syndrome[42] (excess beta-band power), Rett syndrome[43] and Huntington's disease (excess theta-power)[44], in which EEG may be similarly informative for drug development.

We found that four of the five BSID scales and all VABS subdomains showed changes above expectation from NH trajectories, with

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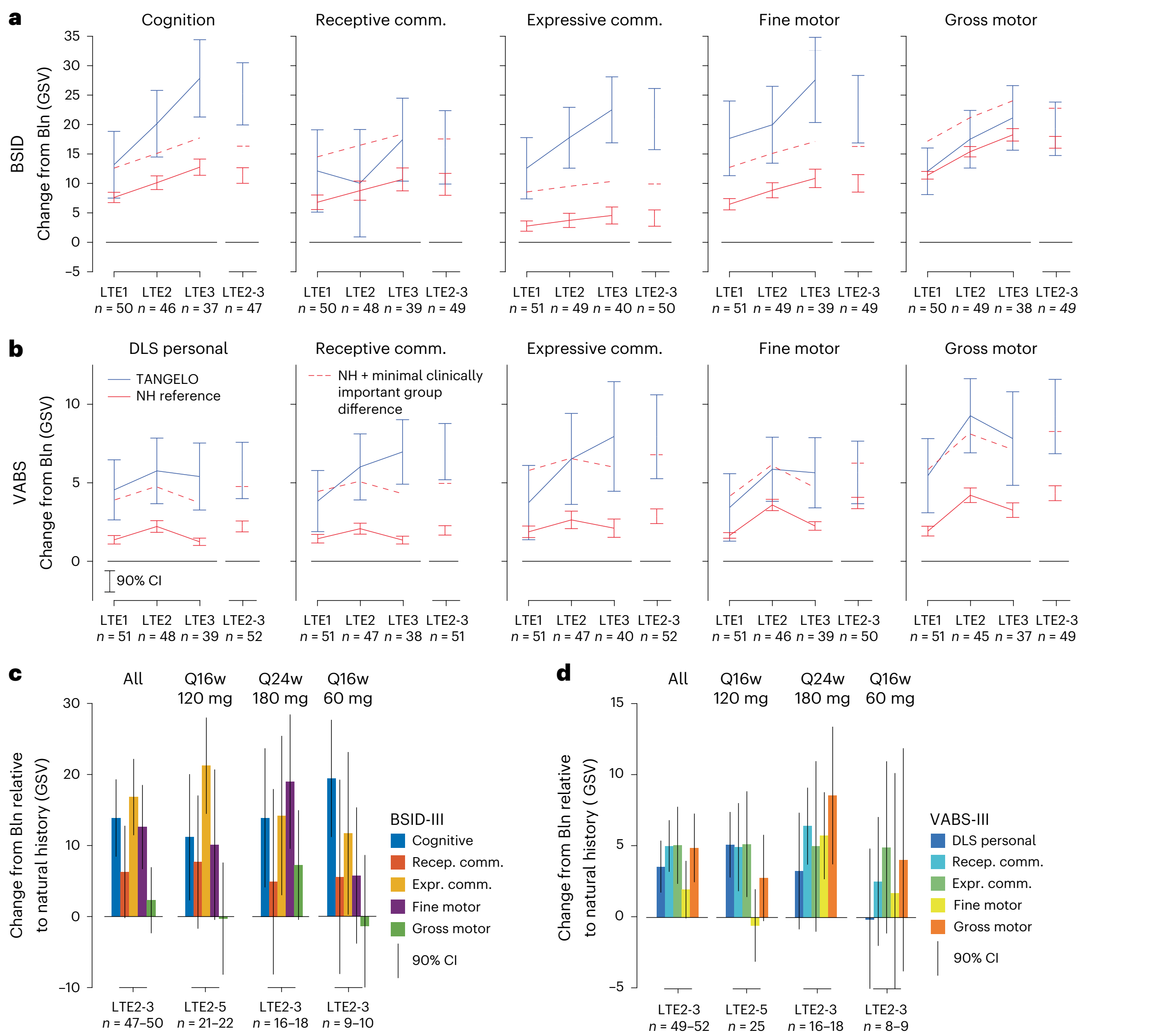


**Fig. 4 | Clinical response in the LTE. a**, BSID: LS mean changes from baseline for all LTE participants (blue) and age-matched NH model prediction (red). The error bars show 90% CIs. The red dashed lines show an estimate of the minimal clinically important group difference on top of the NH model prediction (Supplementary Table 7). Positive changes reflect improvements. **b**, Same as **a** for VABS. **c**, BSID: LS mean changes from NH for all LTE participants and the three different LTE dose regimens and for all BSID domains. **d**, Same as **c** for VABS domains. The error bars show 90% CIs.

differences starting as early as day 100 in the MAD and persisting throughout the LTE. Thus, children gained skills in several key developmental domains at a pace exceeding that of age- and genotype-matched AS children. The magnitude of most changes met or surpassed what we estimated as minimal clinically important group differences (Supplementary Section 8). Because AS is characterized by a profoundly atypical rate of development, including developmental plateauing at about 5–6 years[33], this upward shift in trajectory meets a key treatment goal in AS. On the SAS-CGI (Change for MAD; Severity for LTE), clinicians also reported improvements across clinical domains. The consistency of improvements converging across complementary measures (that is, BSID performance-based assessment completed by the local rater, VABS interview with the caregiver completed by the central rater, SAS-CGI completed by the principal investigator) further substantiates signs of clinical efficacy of rugonersen. In contrast to EEG $\delta$-power, there was no sign of a dose (MAD) or dose regimen (LTE) dependence for the BSID and VABS. However, there was a trend for individuals with a stronger EEG response to show a larger clinical response, which was more pronounced in the LTE, that is, at later timepoints.

The clinical efficacy signals in TANGELO must be interpreted with caution. In an open-label study, treatment-related changes cannot be separated from other trial-related factors such as expectation bias, increased attention to study participants and trial procedures[45]. Naturally occurring developmental changes were accounted for by using NH models. The BSID and VABS, used as endpoints, have limitations in the AS population. The BSID, designed for typically developing children up to 42 months, may not be fully adequate for older children with atypical development. The VABS, while covering a wider age range, may lack sensitivity owing to limited item density in the scale range relevant for AS individuals. Despite these concerns, the BSID, VABS and SAS-CGI results suggest rugonersen-related clinical improvements. However, the TANGELO study design does not allow for a causal assessment of rugonersen's efficacy, which necessitates controlled studies. In addition, *P* values from exploratory analyses in this study should be interpreted as indicative of evidence strength rather than statistical conclusions.

Despite considerable efforts, there are no approved disease-modifying treatments for genetic neurodevelopmental disorders[46].

The genetics of AS offers the opportunity to harness the intact, silenced paternal copy of the disease gene to address the root cause of the disorder. Our study provides clinical data on a therapy targeting UBE3A reinstatement in AS. Our results show that reinstatement of UBE3A in neurons with a *UBE3A*-ATS-targeting ASO can alter pathological brain activity as measured with EEG. The primary analysis of this study shows that rugonersen had an acceptable safety and tolerability profile, supporting further clinical development of rugonersen as a disease-modifying treatment for AS.

## Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41591-025-03784-7.

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[1]Roche Pharma Research and Early Development, Neuroscience and Rare Diseases, Roche Innovation Center Basel, F. Hoffmann-La Roche, Basel, Switzerland. [2]Department of Molecular and Human Genetics, Baylor College of Medicine and Baylor Genetics Laboratory, Texas Children's Hospital, Houston, TX, USA. [3]Department of Pediatrics; Division of Genetics/Dysmorphology, Rady Children's Hospital San Diego, University of California San Diego, San Diego, USA. [4]Roche Pharma Research and Early Development (pRED), Strategy, Portfolio & Operations, Roche Innovation Center Basel, Basel, Switzerland. [5]Department of Neurology and Pediatric Neurology and Erasmus MC center of expertise for neurodevelopmental disorders ENCORE, Erasmus MC, Rotterdam, The Netherlands. [6]NOVA Medical School, Lisbon, Portugal. [7]Roche Products Ltd, Welwyn Garden City, UK. [8]Product Development Neuroscience, Roche Innovation Center Basel, F. Hoffmann-La Roche, Basel, Switzerland. [9]Division of Pediatric, Parc Taulí Hospital Universitari. Institut d'Investigació i Innovació Parc Taulí (I3PT-CERCA), Sabadell, Spain. [10]Roche Pharma Research and Early Development, Pharmaceutical Sciences, Roche Innovation Center Basel, F. Hoffmann-La Roche, Basel, Switzerland. [11]Pediatric Neurology Department, Hospital Sant Joan de Déu, Institut de Recerca Sant Joan de Déu, Barcelona, Spain. [12]UNC Carolina Institute for Developmental Disabilities (CIDD), Dept. of Psychiatry, and Neuroscience Center, University of North Carolina School of Medicine, Chapel Hill, NC, USA. [13]Departments of Pediatrics, Neurological Sciences, Anatomy and Cell Biology, Rush University Medical Center, Chicago, IL, USA. [24]These authors jointly supervised this work: Elizabeth Berry-Kravis, Azad Bonni. *Lists of authors and their affiliations appear at the end of the paper. ✉e-mail: joerg.hipp@roche.com

**the Rugonersen Study Group**

**Muhamed Barakovic[1], Charlotte Bon[10], Azad Bonni[1,24], Ponghatai Boonsimma[1], Ina Bruenig-Traebert[1], Mariana Bustamante[14], Katalin Buzasi[8], Stormy Chamberlain[1], Daniel Chan[4], Susanne Clinch[15], Siobhan Connor-Ahmad[15], Philipp Daetwyler[10], Oana Danila[8], Paulo Fontoura[6], Jörg F. Hipp[1], Sebastian Holst[1], Gregory Hooper[7], Ravi Jagasia[1], Thomas Kremer[1], Michelle L. Krishnan[1], Florian Lipsmeier[14], Stefano Magon[1], Amanda Messer[7], Lorraine Murtagh[1], David Nobbs[14], Alessandro Noci[8], Giuseppe Palermo[8], Dietmar Schwab[10], Lisa Squassante[8], Julian Tillmann[1], Jorrit Tjeertes[8], Brenda Vincenzi[8] & Thomas Wiese[1]**

[14]Roche Pharma Research and Early Development, Data and Analytics, Roche Innovation Center Basel, F. Hoffmann–La Roche Ltd., Basel, Switzerland. [15]Patient Centered Outcomes Research, Roche Products Ltd, Welwyn Garden City, UK. A full list of members and their affiliations appears in the Supplementary Information.

**all TANGELO investigators**

**Carlos A. Bacino[2], Jennifer M. Bain[16], Elizabeth Berry-Kravis[13,24], Lynne M. Bird[3], Jamie Capal[17], Ralitza Gavrilova[18], Ana Roche Martínez[9], María Muñoz Cabeza[19], Nicola Pietrafusa[20], Rajsekar Rajamaran[21], Shafali Jeste[22], Mercedes Serrano[11], Mark D. Shen[12], Federico Vigevano[23] & Marie Claire de Wit[5]**

[16]Columbia University Irving Medical Center, Department of Neurology, Division of Child Neurology, and NewYork Presbyterian Morgan Stanley Children's Hospital of New York, New York, NY, USA. [17]Department of Neurology, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA. [18]Department of Neurology and Department of Clinical Genomics, Mayo Clinic, Rochester, MN, USA. [19]Pediatric Neurology Department, Hospital Virgen del Rocío, Sevilla, Spain. [20]Neurology, Epilepsy and Movement Disorders Unit, Bambino Gesù Children's Hospital, IRCCS, Rome, Italy. [21]UCLA Mattel Children's Hospital, Los Angeles, Division of Pediatric Neurology, University of California, Los Angeles, California, USA. [22]Dept of Pediatrics and Neurology, USC Keck School of Medicine, Children's Hospital Los Angeles, Los Angeles, USA. [23]Head of Developmental Disabilities Dpt IRCCS San Raffaele—Rome, Rome, Italy.

## Methods

### Study design and participants

TANGELO is a first-in-human, open-label, multicenter phase 1 study of rugonersen to investigate the safety, tolerability and pharmacokinetics in children aged 1–12 years with deletion or mutation AS ($n$ = 61; F/M: 28/33; NCT04428281; sponsor: F. Hoffmann-La Roche). The study started in August 2020 and consists of a MAD phase followed by an LTE phase, which have both been completed, and an OOE phase that is ongoing at the time of writing of this paper (Fig. 1). The study is conducted at 12 sites in the United States, Italy, Spain and the Netherlands.

Eligible participants were between 1 and 12 years old, had a body weight of at least 8 kg, were able to tolerate blood draws and undergo lumbar punctures and IT injections (sedation or general anesthesia as indicated by the clinical condition of the participant) and had a confirmed genetic diagnosis of AS (mutation of the maternal *UBE3A* allele or deletion on the maternally inherited chromosome 15q11.2-q13.1 that includes the *UBE3A* gene and is less than 7 Mb in size). Males and females were included with no hypothesis for any sex difference in outcome measures (sex assigned by the clinician). Sex was therefore not a stratification factor and was not a factor in the main analyses.

The MAD investigated 50 participants and was completed in November 2022 when the last participant transitioned to the LTE. Participants were recruited in cohorts of $n$ = 6 (except cohort A2, in which $n$ = 8 was recruited to account for missing key baseline data in two participants; one participant was reassigned from A4 to A3 to align with actual MAD doses administered). This group was chosen as common practice for group sizes in Ph1 studies to assess safety and tolerability. Cohorts comprise two different age groups (A cohorts: 5–12 years; B cohorts: 1–4 years; Table 1). Within each cohort, participants received two or three doses of rugonersen during the treatment period of 8 weeks (individual doses: 6–180 mg; Fig. 1b). For most cohorts, doses were escalated within the cohort, that is, the dose level increased over the two or three doses. The older A cohorts always preceded the younger B cohorts and subsequent A cohorts with higher doses. Before new cohorts were opened, all data were reviewed by an independent board (Study Oversight). The MAD treatment period was followed by a 24-week posttreatment period, during which no drug was administered. The posttreatment period included two in-clinic visits on days 100 and 224 with comprehensive assessments. All participants completed the MAD part of the study.

The MAD was followed by the LTE that started in November 2022 and ended in 23 September 2024. The LTE was designed to investigate the safety and tolerability of rugonersen at stable dose regimens and presented an opportunity for continued treatment for the participants. The start of the LTE was conditioned on completion of the chronic 39-week NHP toxicologic study, which resulted in participants from cohorts A1–A3 and B1–B2 not seamlessly transitioning into the LTE. To bridge the time till the start of the LTE, participants from these cohorts received one bridging dose (BD) following the completion of the MAD posttreatment period (despite the BD, some participants had a dosing gap of up to 9 months). Participants were assigned to one of three LTE cohorts that differed by the dose strength and dosing frequency, that is, dose regimens (Fig. 1b). Participants from MAD cohorts A1, A2, B1 and B2 transitioned to a dose regimen of 120 mg every 16 weeks (Q16w-120mg). Participants from MAD cohorts A3, A4, B3 and B4 transitioned to a dose regimen of 180 mg every 24 weeks (Q24w-180mg). In total, 49 of 50 participants from the MAD continued in the LTE. One participant (cohort A3) decided not to participate in the LTE owing to perceived limited improvements observed in the MAD as shared by the caregiver. A total of 11 new participants were recruited to receive 60 mg every 16 weeks (Q16w-60mg). The 11 participants were considered sufficient to provide safety and tolerability data for this additional dose regimen.

Following the LTE, participants and their caregivers were offered the opportunity to continue into the OOE that started in September 2024 and is still ongoing during the time of writing of this paper (it is planned to be completed in 2025). The OOE has a curtailed schedule of assessments, with the main objective of providing open-label access to rugonersen within the context of a clinical trial to ensure that comprehensive safety assessments and monitoring are conducted while reducing the burden of exploratory assessments.

This publication focuses on the analysis of MAD and LTE data. Analyses with dose as a factor are as-treated analyses; that is, they use the actual doses administered (see more details below). Unless stated otherwise, the data cut for all analyses reported is 23 September 2024. This is the date protocol version 10 became active and the OOE started.

### Study conduct and dropouts

The study was substantially impacted by the sponsor's announcement on 20 June 2023 to discontinue the internal clinical development of rugonersen and seek an external partner to continue the program. In addition, an end date for the study was initially set on 29 February 2024 to facilitate the transition to a new partner. However, no agreement was reached with the potential partner, and the sponsor subsequently announced on 25 April 2024 that rugonersen remained accessible in an OOE phase (as described above) until the second quarter 2025, while the search for a development partner is ongoing.

Up to the point of the first announcement, 1 of 61 participants completed the MAD part of the study and decided not to continue with the LTE. In addition, 3 of the remaining 60 participants discontinued the study during the LTE phase upon the participants' request. Following subsequent announcements, a substantial number of participants either decided to leave the study or completed their final visit. With the introduction of the OOE, the study offered participants who had discontinued the study a chance to return for further treatment. At the time of writing, 28 participants discontinued post-sponsor announcement and did not return to the OOE, and 30 continued and/or restarted the study as part of the OOE (Fig. 1).

### Study drug

The dose of rugonersen (RO7248824) was guided by toxicology studies complemented with a translational pharmacokinetic–pharmacodynamic (PK–PD) model that was based on cynomolgus monkey studies. The monkey PK–PD studies assessed brain kinetics of rugonersen and *Ube3a*-ats suppression, and *Ube3a* mRNA and UBE3A protein increase. The dose used in monkeys was scaled by a factor of 10 to determine the appropriate dose for humans, based on physiological differences between monkeys and humans. The starting dose for A cohorts was selected to have short and low expression of *UBE3A* according to the model (15 mg). The starting dose for B cohorts was selected to be subtherapeutic according to the model (6 mg). Intraparticipant dose escalation was chosen to test a broad dose range. The maximum dose was limited to what was deemed safe in the monkey toxicity study.

### IT administration

IT administration was performed under sedation or general anesthesia via lumbar puncture (LP) using a needle inserted into the L3–L4 space, although placement at a different level was allowed if participant anatomy or clinical judgment dictated. Needles with at least 22 G were required. Before the injection, the CSF opening pressure was measured with a manometer. Following this, a minimum of 5 ml of CSF was collected for analysis. Each dose of rugonersen was administered as a single IT bolus injection of 5 ml over 1–2 min. Following the bolus administration of the drug, a 5-ml intrathecal flush bolus of low-endotoxin diluent solution was given, and participants were kept in the Trendelenburg (head-down) position for 30 min up to 1 h. The use of the flush was discontinued later in the study following data from

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a clinical PET study with radio-labeled rugonersen that did not support a presumed advantage in terms of biodistribution with flush.

### Study oversight

The trial was conducted in accordance with the Declaration of Helsinki. The trial protocol and all documentation were approved by the institutional review board or independent ethics committee at each investigational site (for a list of ethics committees approving the study, see Supplementary Section 1). All caregivers or legal representatives of the participants provided written informed consent. Participants and their families received no compensation. The trial was sponsored by F. Hoffmann-La Roche, which provided the study drug (rugonersen). Personnel from F. Hoffmann-La Roche designed the trial with input from patient representatives, academic investigators and other disease experts. A Dose Decision Committee, composed of sponsor personnel with medical and clinical trial expertise, and external scientific experts, authorized each dose escalation after reviewing safety data. The investigators collected the data, which were held and maintained by the sponsor. Data were analyzed by personnel from the sponsor and were interpreted by all authors. The investigators vouch for the fidelity of the protocol and protocol amendments. We vouch for the completeness and accuracy of the data. We and the sponsor made the decision to submit the paper for publication.

### Clinical safety

To assess adverse event severity, we used the adverse event severity grading scale, the National Cancer Institute Common Terminology Criteria for Adverse Events v5.0.

### Pharmacokinetics

Rugonersen concentrations were determined by a hybridization-based enzyme-linked immunosorbent assay. The hybridization-based enzyme-linked immunosorbent assay was validated to determine the rugonersen concentrations in human plasma and CSF. The lower limit of quantification and calibration range of the assays were 0.0490 nM and 0.0490–3.00 nM, respectively. The inter- and intra-assay precision and accuracy were within the acceptance criteria of 20% (25% at the lower and upper limit of quantification) for each assay. Rugonersen was stable in human plasma for a maximum of 9 days at room temperature and up to eight freeze–thaw cycles. When stored at −20 °C or −70 °C, rugonersen was stable for around ~12 months in human plasma. Rugonersen was stable in human CSF for a maximum of 1 day at room temperature and up to five freeze–thaw cycles. When stored at a nominal 20 °C or 70 °C, rugonersen was stable for ~23 months in human CSF.

The PK properties of rugonersen in plasma were evaluated based on plasma PK data within 24 h of drug administration. No accumulation in plasma after repeated dosing was observed; therefore, the preliminary non-compartmental analysis was conducted independent of dosing frequency. Pharmacokinetic data are available for all 61 participants treated with doses ranging from 6 mg to 180 mg. A fraction of PK (plasma and CSF) and anti-drug antibody (ADA) samples were excluded from the analysis as during shipment, samples thawed for a duration outside of the drug stability window. Only 120-mg and 180-mg dose levels were affected.

The cutoff date for PK analyses was February 2024.

### Collection of EEG data

Awake EEGs of 20-min duration were recorded at clinic visits with an international 10–20-system (19 electrodes) montage. Data were recorded with Grael EEG 4K amplifiers (Compumedics; sampling rate, 512 Hz; low-pass filter cutoff, 0.3 Hz; high-pass filter cutoff, 70 Hz; notch filter, 50/60 Hz (EU/US); reference electrode location, POz; ground electrode location, AFz) with gel electrodes mounted in caps (four different cap sizes were available: newborn, Waveguard ANT, 39–43 cm; infant, Waveguard ANT, 43–47 cm; child, Waveguard ANT, 47–51 cm; and small, Quick Cap Neo Net, Compumedics, 51–57 cm). Impedances were below 20 kΩ. EEG devices were prepared centrally by the EEG vendor and distributed to sites. Tailored EEG acquisition training was provided, and all sites underwent an accreditation process that included performing an independent recording with the equipment, uploading the data to the EEG vendor and passing the quality criteria for quantitative EEG analyses as defined by the central vendor. A total of 639 EEGs were recorded.

### Reading of EEGs

EEGs were reviewed centrally by epileptologists within 72 h, and the readings were forwarded to the medical monitor and the sites.

### Preprocessing and endpoints for quantitative analyses of EEG

For quantitative analyses, EEG data were bandpass filtered (0.5–45 Hz, FIR filter, 2 s) and cleaned of artifacts (interpolation of artifactual channels; exclusion of sections with strong artifacts; removal of independent components (ICs) that capture electrooculogram, electromyogram or technical artifacts) and re-referenced to average across all channels[27,28,34]. This careful preprocessing was done to ensure that signals analyzed reflect brain activity rather than technical artifacts such as eye movements. Preprocessing was performed by trained personnel of the EEG vendor according to a detailed manual. For 5.5% of datasets, electrodes were identified as artifactual and interpolated (one electrode: 4.1%; two to four electrodes, 1.4%; only datasets with maximally two interpolated electrodes were used for quantitative analyses). On average, 5.7 ICs were rejected (s.d., 2.24; quantiles, [4,7]) and 67% of data corresponding to 17.2 min were retained for analyses (s.d., 5.35 min; quantiles, [13.5,20.2] min). A subset of the EEG datasets was preprocessed by two technicians ($n$ = 309), allowing the estimation of inter-artifact rejector reliability. The inter-artifact rejector reliability for the EEG endpoints was very high ($r$ = 0.989 for the main endpoint EEG $\delta$-power); that is, the results are very robust to the artifact rejection process, which has an element of subjectivity.

Power spectral estimates were derived using Morlet wavelets (https://roche.github.io/neuro-meeglet/; parameter bandwidth: bw = 1/3 octave (oct), corresponding to $f/\sigma_f = 10.23$, with $f$, center frequency, and $\sigma_f$, spectral bandwidth)[47,48]. Center frequencies were spaced logarithmically according to the exponentiation of the base 2 with exponents ranging from 0 (1 Hz) to 5.33 (~40 Hz) in steps of 1/12. Power spectral estimates were derived as average power values of successive 3/4-overlapping temporal windows. Absolute power density values were then averaged across electrodes, where applicable, and had a unit of $\mu V^2\ oct^{-1}$.

The main EEG endpoint, $\delta$-power, was derived by integration over the frequency range 2–4 Hz and then averaged across all electrodes (unit, $\mu V^2$). For supplementary analyses, we derived several additional EEG endpoints. Total power was defined as the integral over the whole frequency range investigated (1–40 Hz; unit, $\mu V^2$). Relative $\delta$-power was derived by dividing $\delta$-power by the total power (unitless). Peak $\delta$-power was the power spectral density value at 2.8 Hz (AS population peak frequency), which is an alternative metric used in some of our previous works (unit, $\mu V^2\ oct^{-1}$)[27]. Furthermore, we report in supplementary analyses $\delta$-power and relative $\delta$-power as defined by the International Pharmaco-EEG Society ($\delta$, 1.5–6 Hz; total power to derive relative $\delta$: power, 1.5–30 Hz; based on Fourier transform of 2 s long windows; unit, $\mu V^2$)[49]. For analyses of endpoints, we used $10 \times \log_{10}$ to render distributions more normal, except for relative power estimates. Consequently, differences between such signals have the unit decibel (dB). See Supplementary Sections 12 and 13 for analysis of alternative EEG endpoints.

### Developmental trajectory of EEG endpoints

TANGELO was an open-label study with no control arm. To generate a synthetic control arm, we therefore built models of the AS developmental trajectory, that is, natural change with age, for all EEG endpoints. We used non-drug and NH data from children (1–16 years) with AS and typical developing controls from three sources: AS-NHS[27], FREESIAS[34], and baseline and screening data from TANGELO. This dataset comprised 169 recordings from 120 children with Del AS, 31 recordings from 26 children with Mut AS and 70 recordings from 58 typical developing children. Data were preprocessed identical to TANGELO data (see above).

We followed the approach described in a previous study[27], of fitting linear mixed models of the following structure:

$$\mathrm{EEG} \sim 1 + \log(\mathrm{age}) + \mathrm{Genotype} + \mathrm{Data\ source} + \mathrm{Electrode} + (1|\mathrm{Participant}), \quad (1)$$

where 'EEG' is the EEG endpoint of interest (for example, log-transformed delta-band power averaged across all electrodes), 'log(age)' is log-transformed age ($\log_2$) centered to 0 by subtracting the mean across all recordings, 'Genotype' (Del, UBE3A) is deletion AS or *UBE3A* mutations (is skipped for TD models, which were fit independently), 'Data source' (AS-NHS, FREESIAS, TANGELO) is used to account for potential differences in EEG amplifiers used across studies, 'Electrode' refers to electrodes used in a model (for example, all 10–20 system electrodes) and was skipped if and when the average is modeled and '(1|Participant)' is a random effect for participants to account for repeated measures. Note that there were no sufficient longitudinal data to model a slope.

Models were fit independently for AS and controls. To capture the confidence in the model fit, we performed bootstrapping ($n$ = 1,000). These bootstraps were used for illustration and to estimate the variability when simulating the synthetic control arm for the TANGELO analyses (Supplementary Fig. 2 and Supplementary Table 21 in a separate file containing the age–EEG $\delta$-power relationship for Del and Mut AS).

### Selection of EEG data for analyses

In total, 639 EEGs were recorded in different study parts (MAD, BD, LTE; 18 were unscheduled at the discretion of the sites, for example, to follow up after a period with seizures; 61 were planned but not performed, and of these, one failed owing to technical issues; 17 were in the context of early termination; for the others, no reason was recorded; one EEG was discarded from analysis owing to poor signal quality) at varying doses and dosing intervals. We performed two sets of analyses, one focusing on the MAD part up to the day 224 and one focusing on the LTE part. Analyses incorporating all data, for example, using pharmacodynamic–pharmacokinetic or kinetic–pharmacodynamic modeling, are beyond the scope of this paper.

For the MAD analyses, we excluded data from two participants (one participant had no baseline EEG, and the other had a substantial delay of the second dose and subsequent visits were out of phase). Of the remaining 48 participants, we used 264 datasets from up to 6 visits and the baseline visit (Table 1). Datapoints were excluded if they were not within ±7 days of the assessment day as indicated in the protocol, measured from the first MAD dose for visits on days 28 and 55, and ±14 days as measured from the last MAD dose for all other days (timing of included datapoints: day 28: 28.0 ± 1.6 days (mean ± s.d.), 26–35 days (range); day 55: 56.2 ± 1.7 days, 52–63 days; day 70: 69.9 ± 2.9 days, 63–76 days; day 100: 99.6 ± 4.6 days, 86–111 days; day 140: 138.7 ± 5.6 days, 127–147 days; day 224: 224.0 ± 8.7 days, 202–250 days).

For the LTE analyses, five participants did not provide data or were excluded (one participant had no baseline EEG, one participant received a different dose as decided by the PI and two participants terminated early in the LTE and therefore did not provide data) and one participant chose not to transition from the MAD to the LTE. In total, we used 236 datapoints from 56 participants (11 new participants; 45 transitioning from the MAD) for the analysis of up to 6 LTE visits and the baseline visit. Participants were assigned to one of three dose regimens ('Study design and participants'). One participant received a different dose regimen as originally planned and was reassigned accordingly. The B4 cohort received 120 mg and 180 mg 2 months apart in the MAD and then continued with Q24w-180mg. We defined the last MAD dose as the first LTE dose. All other cohorts had either no dose before the first LTE dose (EA1/EB1) or the last dose was at least 6 months apart. Given the heterogeneity in dosing before the LTE start, we refrained from analyzing the EEG before the first LTE dose. We used EEG data only if the correct dose levels were given up to this point (that is, we stopped using further participant data if the dose given was different than the one assigned) and used data only if the last dose given was ±28 days around the expected time (16 weeks or 24 weeks, depending on the cohort) and if the second to the last dose was ±56 days around the expected time (32 weeks or 48 weeks, depending on the cohort). Due to the study design with early MAD cohorts transitioning to EA2/EB2, late MAD cohorts and transitioning to EA3/EB3 (and with a larger dose interval), and new cohorts transitioning to EA1/EB1, there are different numbers of visits by dose regimen (Q16w-120mg: LTE1–6; Q16w-60mg: LTE1–3; Q24w-180mg: LTE1–3).

### Statistical analysis of EEG data

**Analysis of MAD data.** For the analysis of EEG endpoints, we used linear models that incorporate an age- and genotype-matched NH prediction of change and with an intercept, cumulative dose from day 1 up to the point the EEG was taken, the baseline deviation from the NH prediction and log(age) as covariates.

Models were fit separately for timepoints available for all participants (days 55, 100 and 224; days 28, 70 and 140 were only available for a subset and are reported in Supplementary Table 10).

$$y^{\Delta_{\mathrm{NH}}}{}_{T} \sim 1 + \mathrm{CumDose} + y^{\Delta_{\mathrm{NH}}}{}_{\mathrm{Bln}}\,(\perp \mathrm{CumDose}) + \log(\mathrm{age})\,(\perp \mathrm{CumDose}) \quad (2)$$

where $y_T$, $y_{\mathrm{Bln}}$ are the EEG parameter of interest at the timepoint $T$ and at Bln (for example, log-transformed delta-band power averaged across all electrodes at day 100 and at baseline, respectively) and $y^{\Delta \mathrm{NH}}$ indicates the deviation of $y$ from the NH model prediction (see above for the description of the NH model) for matched age and genotype ($y^{\Delta_{\mathrm{NH}}}(\mathrm{age}, \mathrm{genotype}) = y_{\mathrm{measured}}(\mathrm{age}, \mathrm{genotype}) - y_{\mathrm{NH-Prediction}}(\mathrm{age}, \mathrm{genotype})$). Using $y^{\Delta_{\mathrm{NH}}}$ instead of $y$ accounts for expected changes over the course of the study, owing to maturation (Supplementary Fig. 2b). This correction takes the nonlinear relationship between age and natural change in EEG parameters into account and therefore adequately prepares the data for use in the linear model. 'log(age)' is the $\log_2$ of the age of participants at the visit, 'CumDose' is the cumulative dose that a participant received up to the timepoint $T$ of interest in mg, 'Var(⊥CumDose)' indicates that a predictor variable ('log(age)' or $y^{\Delta_{\mathrm{NH}}}{}_{\mathrm{Bin}}$) is orthogonalized with respect to CumDose; that is, the linearly predictable part is subtracted. We performed the orthogonalization to minimize that chance correlations between CumDose and $y^{\Delta_{\mathrm{NH}}}{}_{\mathrm{Bin}}$ in this small sample would influence the fit of the CumDose predictor. For log(age), the situation is more complicated. By design, lower doses were given for younger participants (Fig. 1b). We used $\log(\mathrm{age})\,(\perp \mathrm{CumDose})$ and performed sensitivity analyses (Supplementary Section 12) to increase confidence that CumDose effects are not explained by age effects.

All covariates (CumDose, $y^{\Delta_{\mathrm{NH}}}{}_{\mathrm{Bin}}$, log(age)) were mean centered before being subjected to the model and before orthogonalization. We report least squares (LS) means throughout the paper. Fitting models separately for each timepoint avoided assumptions such as, for example, missing at random.

See Supplementary Section 12 for additional analyses testing alternative models, investigation of subpopulations and alternative EEG endpoints, and sensitivity analyses.

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**Analysis of the LTE data.** For the analysis of EEG endpoints in the LTE, we used linear models that incorporate an age- and genotype-matched NH prediction of change and with an intercept, the baseline deviation from the NH prediction and log(age) as covariates.

Models were fit separately for each dose regimen and LTE visit (Q16w-120mg, Q24w-180mg, Q16w-60mg). For the definition of covariates, see 'Analysis of MAD data'.

$$y^{\Delta_{\mathrm{NH}}}{}_{T} \sim 1 + y^{\Delta_{\mathrm{NH}}}{}_{\mathrm{Bln}} + \log(\mathrm{age}) \quad (3)$$

See Supplementary Section 13 for additional analyses testing alternative models, investigation of subpopulations and alternative EEG endpoints, and sensitivity analyses.

## Measures of clinical efficacy

To assess the exploratory efficacy of rugonersen on core symptoms of AS, including cognition, receptive and expressive communication, and fine and gross motor function, we used three complementary instruments: the BSID[30], an individually administered performance outcome assessment; the VABS[31], a semi-structured interview with the caregiver; and the SAS-CGI[32], a clinician-rated assessment of severity (SAS-CGI-Severity) and symptom change (SAS-CGI-Change). Data from multiple observational and clinical research studies[33,34,50,51] support the validity and sensitivity of the BSID and VABS to assess the most relevant symptoms in AS[6]. Furthermore, NH data showed that the BSID and VABS can track development in AS and differentiate between AS genotypes[33], suggesting that these instruments are sufficiently sensitive to detect treatment response.

**BSID.** The BSID[30] is a performance outcome assessment used to quantify the developmental functioning and developmental delay in infants and young children. A trained rater presents in a structured and standardized testing session various developmental tasks to generate an observable behavioral response across cognitive, language (expressive, receptive) and motor (fine, gross) domains. Although the BSID is normed for individuals up to 42 months of age, it is widely used beyond the normative age cutoff to assess developmental skills in older children with severe–profound levels of cognitive impairment, including AS[33,34,50,51]. This is primarily because individuals with severe disabilities are better assessed by standardized instruments appropriate to their developmental, rather than chronological, age. We used growth-scale values (GSVs) as the main exploratory clinical endpoint, which were created by fitting a Rasch model to the item-level data of the standardization sample for each domain[52]. GSVs have favorable properties for measuring change in clinical trials including an equal-interval scale and same mean and standard deviation across subtests. A higher score indicates better developmental skills.

**VABS.** The VABS[31] is a semi-structured interview that measures an individual's adaptive behavior across three domains of communication, socialization and daily living skills. Additional domains that can be assessed include motor skills (fine and gross motor skills) and maladaptive behaviors. Each adaptive behavior domain is composed of three subdomains. The following five subdomains were analyzed: communication (receptive, expressive), daily living skills (personal) and motor skills (gross and fine). Due to floor effects, the following subdomains were not analyzed: socialization (interpersonal relationships, play and leisure time, coping skills) and communication (written). Subdomain raw scores are used to derive GSVs. GSVs are person-ability scores that are used to track an individual's progress. A higher score indicates better adaptive functioning.

**SAS-CGI.** The SAS-CGI[32] was designed to assess, from a clinician's perspective, the severity of core symptoms associated with AS and overall, as well as change over time (for example, over the course of a clinical trial). It was developed consistent with the FDA Patient Focused Drug Development Guidance 3 (FDA 2022), whereby interviews with caregivers and clinicians were used to develop a patient-focused conceptual model[6], which informed the domains included in the SAS-CGI. This resulted in nine domains: (1) communication, (2) cognitive ability, (3) fine motor, (4) gross motor, (5) sleep, (6) seizures, (7) inappropriate behaviors, (8) daily living and (9) the domain that assesses overall AS symptom severity. There are two types of SAS-CGI, one for measuring severity (SAS-CGI-Severity) and one for measuring change (SAS-CGI-Change). For the SAS-CGI-Change, clinicians rate the participants' clinical change compared with symptoms at baseline using a 7-point scale, ranging from very much improved (1) to very much worse (7). For the SAS-CGI-Severity, clinicians rate the participants' disease severity using a 6-point scale, ranging from none (0) to very severe (5). We used the SAS-CGI-Change to assess changes in the MAD. For the LTE, we opted to analyze the SAS-CGI-Severity as the baseline reference that was used to assess change for the SAS-CGI-Change was up to 4 years old.

## Developmental trajectories of clinical scales

TANGELO was an open-label study with no control arm. Therefore, to generate a synthetic control arm, we built a model of the AS developmental trajectory (as for EEG, see above). We used non-drug and NH data of individuals with AS from four sources: AS-NHS[33], new AS-NHS (NCT04507997), FREESIAS[34], and baseline and screening data from TANGELO. Combined, there were 668 assessments from 248 individuals with Del AS and 89 assessments from 39 individuals with Mut AS for BSID, and 232 assessments from 163 individuals with Del AS and 79 assessments from 44 individuals with Mut AS for VABS. For a few assessments, some subscales are missing, leading to slightly different numbers across domains.

We followed the approach described in a previous study[33] of fitting linear mixed models of the following structure:

$$\mathrm{Scale} \sim 1 + \log(\mathrm{age}) \times \mathrm{Genotype} + \log(\mathrm{age})^2 \times \mathrm{Genotype} + \log(\mathrm{age})^3 \times \mathrm{Genotype} + \mathrm{Dataset} + (1 + \log(\mathrm{age})|\mathrm{Participant}), \quad (4)$$

where 'Scale' is the clinical scale domain to be modeled (for example, GSVs of the BSID cognitive scale); log(age), $\log(\mathrm{age})^2$, $\log(\mathrm{age})^3$ are third-order orthogonal polynomials of centered $\log_2(\mathrm{age})$; 'Genotype' is the categorical predictor of genotype (Del, UBE3A); and 'Dataset' (AS-NHS, FREESIAS, TANGELO_BLN) is fixed effect for sites. For AS-NHS, the largest dataset, separate fixed effects were used for the six sites participating in this study. (1 + log(age)|Participant) is the intercept and slope random effects for participants to account for repeated measures. For VABS subdomains, we fitted only an intercept as no sufficient longitudinal data were available to estimate a random effect for slope.

To capture the confidence in the model fit, we performed bootstrapping ($n = 1{,}000$). These bootstraps were used for illustration and to estimate the variability when simulating the synthetic control arm for the TANGELO analyses (Supplementary Fig. 3 and Supplementary Table 22 containing the age–clinical scales relationships for Del and Mut AS).

## Selection of clinical scale data for analyses

In total, 442 datapoints from 59 participants for BSID, 458 datapoints from 61 participants for VABS, 572 datapoints from 61 participants for SAS-CGI-Severity and 440 datapoints from 60 participants for SAS-CGI-Change (SAS-CGIs were taken more frequently than BSID and VABS; only data coinciding with BSID and VABS were used here) were collected in different study parts (MAD, bridging doses, LTE) and passed general QC criteria. We performed two sets of analyses, one focusing on the MAD part up to the day 224 and one focusing on the LTE part.

For the MAD, we analyzed data from 135 datapoints from 48 participants for BSID and 128 datapoints from 49 participants for VABS from two visits (day 100, day 224) and the baseline visit for VABS and BSID (Table 1). For the SAS-CGI-Change, we analyzed 90 datapoints from 49 participants from two visits (day 100, day 224). Datapoints were excluded if they were not within ±14 days from the planned day as measured from the last MAD dose (timing of included datapoints: BSID: day 100: 99.2 ± 4.7 days, 86–111 days; day 224: 223.9 ± 6.2 days, 210–244 days; VABS: day 100: 100.3 ± 5.8 days, 86–112 days; day 224: 224.6 ± 5.6 days, 210–238 days; SAS-CGI-Change: day 100: 99.7 ± 4.8 days, 86–111 days; day 224: 224.6 ± 5.3 days, 215–245 days).

For the LTE, we analyzed data from 228 datapoints from 53 participants for BSID, 226 datapoints from 53 participants for VABS and 231 datapoints from 53 participants for the SAS-CGI-Severity. Participants were assigned to one of three dose regimens ('Study design and participants'). One participant received a different dose regimen as originally planned and was reassigned accordingly. The B4 cohort received 120 mg and 180 mg 2 months apart in the MAD and then continued Q24w-180mg. We defined the last MAD dose as the first LTE dose. All other cohorts had either no dose before the first LTE dose (EA1/EB1) or the last dose was at least 6 months apart. Given the heterogeneity in dosing before the LTE start, we refrained from analyzing the data from before the first LTE dose.

We used data only if the correct dose levels were given up to this point (that is, we stopped using further participant data if the dose given was different than assigned) and used data only if the last dose given was ±28 days around the expected time (16 weeks or 24 weeks, depending on the cohort) and if the second to the last dose was ±56 days around the expected time (32 weeks or 48 weeks, depending on the cohort). These rules ensured that data analyzed were in line with the assigned dose regimens.

Due to the study design with early MAD cohorts transitioning to EA2/EB2, late MAD cohorts and transitioning to EA3/EB3 (and with a larger dose interval), and new cohorts transitioning to EA1/EB1, there are different numbers of visits by dose regimen (Q16w-120mg: LTE1–5; Q16w-60mg: LTE1–3; Q24w-180mg: LTE1–3).

### Statistical analysis of clinical scale data

**Analysis of MAD and LTE data.** The base model is a linear model that incorporates an age- and genotype-matched NH prediction of change and with an intercept and the baseline deviation from the NH mean as predictor variables. Models were fitted separately for the two MAD timepoints (days 100 and 224) and for LTE dose regimens (Q16w-120mg; Q24w-180mg; Q16w-60mg).

$$y^{\Delta_{\mathrm{NH}}}{}_T \sim 1 + y^{\Delta_{\mathrm{NH}}}{}_{\mathrm{Bln}} \tag{5}$$

where $y_T$, $y_{\mathrm{Bln}}$ are the clinical scale values of interest at timepoint $T$ and at Bln (for example, growth scores of the BSID cognitive score at day 100 and at baseline) and $y^{\Delta_{\mathrm{NH}}}$ indicates the deviation of $y$ from the NH model prediction (see above for the NH model) for matched age and genotype ($y^{\Delta_{\mathrm{NH}}}(\mathrm{age, genotype}) = y_{\mathrm{measured}}(\mathrm{age, genotype}) - y_{\mathrm{NH-Prediction}}(\mathrm{age, genotype})$). Using $y^{\Delta_{\mathrm{NH}}}$ instead of $y$ accounts for expected changes over the course of the study duration due to maturation of participants. This correction is age dependent, owing to the nonlinear relationship between age and natural change in clinical scores. By using $y^{\Delta_{\mathrm{NH}}}$ instead of $y$, this nonlinearity is accounted for before the data are subjected to the linear model (Supplementary Fig. 4).

The predictor variable ($y^{\Delta_{\mathrm{NH}}}{}_{\mathrm{Bin}}$) was mean centered before being subjected to the model. We report LS means throughout the paper.

See Supplementary Sections 14 and 15 for additional analyses testing alternative models, investigated subpopulations and performed sensitivity analyses.

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

## Data availability

Qualified researchers may request access to individual patient-level data through the clinical study data request platform (https://vivli.org/). Further details on Roche's criteria for eligible studies are available at https://vivli.org/members/ourmembers/. For further details on Roche's Global Policy on the Sharing of Clinical Information and how to request access to related clinical study documents, see https://www.roche.com/research_and_development/who_we_are_how_we_work/clinical_trials/our_commitment_to_data_sharing.htm.

## Code availability

For the collection of EEG data, Curry 8 (Compumedics) was used. For data analyses, R (version 4.2.1, https://www.r-project.org/) and Matlab (version R2024b, https://matlab.mathworks.com/) were used. For EEG analyses, the Matlab FieldTrip toolbox[53] (version 7ba023993, https://www.fieldtriptoolbox.org/) and the MEGLET code[47,48] (https://roche.github.io/neuro-meeglet/) were used.

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## Author contributions

Initial study design and protocol development: M.L.K., D.S., J.F.H., I.B.T., G.H. and J. Tjeertes. Protocol amendments: J. Tjeertes, B.V., D.S., J.F.H., A.N., G.H., I.B.T. and the TANGELO Investigators. Study conduct and data acquisition: B.V., J. Tjeertes, D.C., D.S., J.F.H., G.H., L.M.B., M.C.d.W., M.S., M.D.S., A.R.M., E.B.-K. and the TANGELO Investigators. Data analysis: J.F.H. (EEG and clinical scales), G.H. (safety) and D.S. (PK). Initial paper writing: J.F.H. with support from G.H., D.S., J. Tillmann and A.N. Interpretation of data and final review: all authors.

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## Competing interests

J.F.H., D.C., G.H., L.M., A.N., D.S., J. Tillmann and A.B. are employees of F. Hoffmann-La Roche. I.B.T., M.L.K., J. Tjeertes, B.V., P.F. and R.J. are former employees of F. Hoffmann-La Roche. M.L.K. is currently an employee of Biomedical Research, Translational Medicine, Novartis AG, Basel, Switzerland. J. Tjeertes is currently an employee of Bayer AG, Basel Switzerland. P.F. is Chief Scientific and Clinical Development Advisor, Stalicla SA, Geneva, Switzerland. L.M.B. has received funding from Aardvark Therapeutics, Acadia Pharmaceuticals, Biogen, Ionis Pharmaceuticals, F. Hoffman-La Roche, Gedeon Richter, Neuren Pharmaceuticals, Ovid Therapeutics, PTC Therapeutics, Radius Health, Soleno Therapeutics and Ultragenyx to conduct clinical trials or consult on trial design or results. E.B.-K. has received funding from Acadia, Alcobra, AMO, Asuragen, Avexis, Biogen, BioMarin, Cydan, Engrail, Erydel, Fulcrum, GeneTx, GW, Healx, Ionis, Jaguar, Kisbee, Lumos, Marinus, Mazhi, Moment Biosciences, Neuren, Neurogene, Neurotrope, Novartis, Orphazyme/Kempharm/Zevra, Ovid, PTC Therapeutics, Retrophin, Roche, Seaside Therapeutics, Taysha, Tetra, Ultragenyx, Yamo, Zynerba and Vtesse/Sucampo/Mallinckrodt Pharmaceuticals, to consult on trial design or run clinical or lab validation trials in genetic neurodevelopmental or neurodegenerative disorders, all of which are directed to RUMC in support of rare disease programs. E.B.-K. receives no personal funds, and Rush University Medical Center has no relevant financial interest in any of the commercial entities listed. For M.C.d.W., Erasmus MC received compensation for participation in the Tangelo trial from Hoffmann-La Roche, and for consultation from Jazz Pharmaceuticals. M.D.S. has received research funding from Biogen, Ionis and Roche to conduct clinical trials or research studies. All funds were paid to the University of North Carolina; M.D.S. receives no personal funds and has no relevant financial interest in any of the commercial entities listed. The other authors declare no competing interests.

## Additional information

**Supplementary information** The online version contains supplementary material available at https://doi.org/10.1038/s41591-025-03784-7.

**Correspondence and requests for materials** should be addressed to Jörg F. Hipp.

**Peer review information** *Nature Medicine* thanks Craig Erickson and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Jerome Staal, in collaboration with the *Nature Medicine* team.

# nature portfolio

Corresponding author(s): Jörg F. Hipp

Last updated by author(s): May 8, 2025

# Reporting Summary

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

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| ☐ | ☒ | The exact sample size ($n$) for each experimental group/condition, given as a discrete number and unit of measurement |
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| ☐ | ☒ | A description of all covariates tested |
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| ☐ | ☒ | A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ | For null hypothesis testing, the test statistic (e.g. $F$, $t$, $r$) with confidence intervals, effect sizes, degrees of freedom and $P$ value noted<br>*Give P values as exact values whenever suitable.* |
| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ | Estimates of effect sizes (e.g. Cohen's $d$, Pearson's $r$), indicating how they were calculated |

*Our web collection on statistics for biologists contains articles on many of the points above.*

## Software and code

Policy information about availability of computer code

| | |
|---|---|
| Data collection | For collection of EEG data: Curry 8 (Compumedics). |
| Data analysis | For data analyses R 4.2.1 (https://www.r-project.org/) and Matlab R2024b (https://matlab.mathworks.com/) were used. For EEG analyses the FieldTrip toolbox (https://www.fieldtriptoolbox.org/, version: 7ba023993) and the MEGLET code (https://roche.github.io/neuro-meeglet/) were used. |

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Information and how to request access to related clinical study documents, see: https://www.roche.com/research_and_development/who_we_are_how_we_work/clinical_trials/our_commitment_to_data_sharing.htm.

## Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

| | |
|---|---|
| Reporting on sex and gender | Sex information was collected at screening (assigned by clinician). Sex ratios are reported in the manuscript. Sex was not specifically analyzed, since there was no hypothesis that sex would have an influence on the outcomes. |
| Population characteristics | Covariate-relevant population characteristics: Genotype (Deletion of 15q11.2-13.1, Mutations of UBE3A), age, baseline values of clinical scales and EEG delta power, cumulative MAD dose, EEG delta-power effect on MAD day 100 and LTE2. For natural history data additional covariates were used: Dataset, orthogonal polynomials up to order three of log(age), EEG electrode location (categorical). |
| Recruitment | Participants were recruited by the participating clinical centers according to study eligibility criteria.<br>In total n=61 participants (Sex, F/M: 28/33) were recruited.<br>All participants' caregiver or legal representative provided written informed consent.<br>Participants and their families received no compensation.<br>The sponsor is unaware of any patient selection bias. |
| Ethics oversight | Principal Investigator, IRB/Ethic Committee and Reference number:<br>Dr Maria Muñoz, CEIC Fundació Sant Joan de Dewu, Servei de Recerca, AC-03-20<br>Dr Mercedes Serrano, CEIC Fundació Sant Joan de Dewu, Servei de Recerca, AC-03-20<br>Dr Ana Roche Martinez, CEIC Fundació Sant Joan de Dewu, Servei de Recerca, AC-03-20<br>Dr Marie-Claire de Wit, Central Committee on Research Involving Human Subjects (CCMO), NL73321.000.20<br>Dr Nicola Pietrafusa / Dr Federico Vigevano, Comitato Etico dell' IRCCS Ospedale Pediatrico Bambino Gesù, 2084 / 2020<br>Dr Jennifer Bain, Columbia University Medical Center, CUMC IRB, AAAS9480<br>Dr Ralitza Gavrilova, Mayo Clinic Institutional Review Board Rochester, 20-007499<br>Dr Elizabeth Berry-Kravis, Rush University Medical Center, Rush University Medical Center Institutional Review Board, 00000482<br>Dr Lynne M. Bird, UCSD Human Research Protections Program, Altman Clinical and Translational Institute, 200683<br>Dr Rajsekar Rajaraman / Dr Shafali Jeste, Advarra, 00041741<br>Dr Carlos Bacino, Baylor College of Medicine Office of Research, Institutional Review Board, H-47303<br>Dr Mark Shen / Dr Jamie Capal, Advarra, 00041741 |

Note that full information on the approval of the study protocol must also be provided in the manuscript.

# Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

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# Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

| | |
|---|---|
| Sample size | For the multiple ascending dose (MAD) part groups of n=6 were recruited. This is a common group size in a Ph1 study to assess safety and tolerability. Additional n=11 participants were recruited to for the long-term extension (LTE) phase of the study with the aim to collect additional data on a lower dose (60 mg). Eleven participants were considered sufficient to provide safety and tolerability data to be compared to the other two LTE groups comprised of participants transitioning from the MAD. |
| Data exclusions | Few datapoints were only excluded due to not passing QC. E.g. few EEG datasets that were not deemed suitable for quantitative analyses. |
| Replication | Baseline values (EEG and clinical scales) fell well within the range expected from natural history data (reported in the manuscript). This replicates the EEG phenotype and clinical impairment in the studied population. The results related to drug effects need to be replicated in future studies. |
| Randomization | The study was an open label study. Recruitment into the cohorts was performed according to available participants at clinical sites at each stage of the study. A cohorts recruited 5-12 years old and B cohorts 1-4 years old participants. Stratification factors were genotype (Deletion of 15q11.2-13.1, Mutations of UBE3A) and age (A-cohorts: 5 - 12 years; B-cohorts: 1-4 years). |
| Blinding | Not relevant since open label Ph1 study. |

# Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

## Materials & experimental systems

| n/a | Involved in the study |
|---|---|
| ☒ | ☐ Antibodies |
| ☒ | ☐ Eukaryotic cell lines |
| ☒ | ☐ Palaeontology and archaeology |
| ☒ | ☐ Animals and other organisms |
| ☐ | ☒ Clinical data |
| ☒ | ☐ Dual use research of concern |

## Methods

| n/a | Involved in the study |
|---|---|
| ☒ | ☐ ChIP-seq |
| ☒ | ☐ Flow cytometry |
| ☒ | ☐ MRI-based neuroimaging |

## Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration

NCT04428281

Study protocol

The full study protocol has not been made public. Roche will provide the study protocol upon approval of a research proposal per the data availability statement.

Data collection

Data collection started in August 2020 it is still ongoing in the optional open-label extension (OOE) part of the study (n=61 participants). The clinical data were entered in the Medidata Rave system; the Cognitive scales were entered via a Clinical Ink table; the EEG data was collected using identical setups provided to the clinical sites through a central EEG vendor. Data was by clinical site staff at 12 sites in the US, Italy, Spain and the Netherlands:
Ospedale Pediatrico Bambino Gesu, Italy; PIs: Dr Vigevano (past), Dr Pietrafusa (current)
Erasmus MC - Erasmus Medisch Centrum, Netherlands; PI: Dr de Wit
H. Sant Joan de Deu, Barcelona, Spain; PI: Dr Serrano Gimare
Hospital Virgen del Rocio, Sevilla, Spain; PI: Dr Muñoz
Parc Taulí, Sabadell, Spain; PI: Dr Roche Martínez
RUSH Children's Hospital, United States; PI: Dr Berry-Kravis
UNC School of Medicine - Carolina Institute for Developmental Disabilities, United States; PI: Dr Capal (past), Dr Shen (current)
Texas Children's Hospital, United States; PI: Dr Bacino
University Of California, Los Angeles, United States; PIs: Dr Rajamaran (current), Dr Jeste (past)
Rady Children's Hospital - San Diego, United States; PI: Dr Bird
Columbia University Medical Center, United States; PI: Dr Bain
Mayo Clinic, United States; PI: Dr Gavrilova

Outcomes

From the protocol

Primary: To assess the safety and tolerability profile of RO7248824, the following endpoints were planned to collect and analyzed:
- Frequency and severity of adverse events, serious adverse events, treatment discontinuations due to adverse events.
- Frequency of abnormal laboratory findings (blood and cerebrospinal fluid [CSF]).
- Frequency of abnormal vital signs and ECG values.
- Mean changes from baseline in vital signs (temperature, systolic and diastolic blood pressure, heartrate, respiratory rate) over time.

The results are reported as adverse events and serious adverse events (Table 1).

Secondary: To investigate the plasma pharmacokinetics (PK) of RO7248824, the following parameters were planned to be analyzed:
- Time to maximum concentration (Tmax)
- Maximum plasma concentration observed (Cmax)
- AUC from Time 0 to time of last sampling point or last quantifiable sample, whichever comes first (AUClast), AUC from Time 0 to infinity (AUCinf)

The results are reported in Supplementary Section 5.